

FACULTY OF ENGINEERING

INGB417 - FACILITIES DESIGN

GA Assignment

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1. Background

TIPTOP TABLES has 25 years of outdoor furniture manufacturing experience and has been catering to the local South African Market through unique design ideas that promotes efficiency and customer convenience. The products sold by TIPTOP TABLES includes an outdoor set that is composed of a two-in-one aluminium table, and four aluminium chairs with a rattan covering. The two-in-one table feature enables the table size to be extended or compact depending on the table configuration.

The aluminium patio table is water-resistant and features a charcoal grey protective powder paint coat. The table is lightweight, with a slanted surface for easy cleaning, and easily disassembled by removing the legs for easy storage when not in use. The outdoor table set is the only product line manufactured by TIPTOP TABLES, and demand for the product has increased over the years. As such, the existing manufacturing facility cannot fulfil the anticipated 5000 outdoor sets monthly demand. The increased demand is

TIPTOP TABLES management has contracted the expertise of an Industrial Engineer to enable them to make strategic decisions of the location, size, and facility layout of the new manufacturing facility of the company. Additionally, the managing structure anticipates a 20% increase in the near future; therefore, facility expansion shall be incorporated as part of the systematic engineering design approach.

The purpose of the facility design report is to communicate the systematic design approaches followed, by means of demand calculations, analysing the manufacturing process, conducting equipment research, evaluating alternative design approaches, and delivering a final master facility plan.

2. Design Objectives

The value of designing a new facility for TIPTOP TABLES will enable the company to meet the increased consumer demand and potentially increase the revenue of the company. The facility design offers the opportunity to incorporate refined workflow and material flow to processes.

AIM

Design a manufacturing facility for TIPTOP TABLES that meets customer demand, provides efficient receiving of materials and product dispatch, and facilitates future company expansion.

OBJECTIVES

- Calculate the production capacity to meet consumer demand.
- Determine the space requirements for an effective facility layout.
- Design a facility that shall meet current consumer demand and includes a 20% space expansion based on the forecast provided by management.
- Incorporate regulatory, and safety standards to the facility design.
- Design spaces that are ergonomic to promote employee efficiency, and customer comfort.
- Integrate an effective product distribution system that connects to the public road, which shall be easily accessible by both trucks and customers.

3. Design Requirements

The design requirements for the new TIPTOP TABLES manufacturing facility includes the design details that were adhered to as provided by management.

PRODUCT MANUFACTURING

The different processes have designated workstations that are designed to reduce operator and material movement requirements, which includes the staging of materials and components at the assembly stations.

The automated welding machines, and manual welding workstations have separated spaces as a safety precaution.

STORAGE FACILITIES

The storage requirements of the facility include the area for storing raw material stock and finished products inventory stock. The details of the 3D-space storage requirements are further discussed in section 6 (Detail Design Specifications).

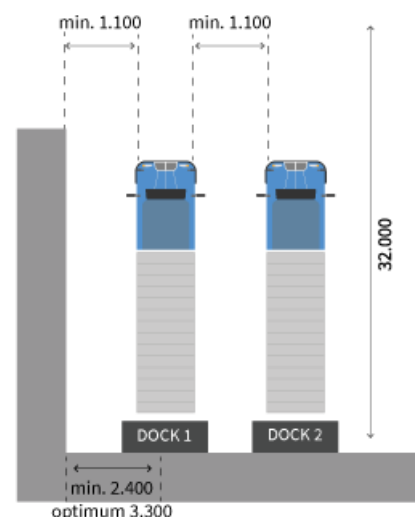
The work in progress (WIP) products are assigned storage carts that are easily moved between processes.

WASTE MANAGEMENT

The facility is regularly cleaned and maintained by the operators to comply with health and safety regulations. Additionally, the discarding of scrap material produced during the cutting process is collected via a scrap bin. The placement of the scrap bin within the facility has been designed to accommodate easy scrap bin dispatch to remove the waste material from the premises.

LOADING BAY

The product distribution system enables suppliers to deliver the raw materials and components throughout the week by means of a loading bay. The same loading bay in Figure 1 is utilised for product dispatch of bulk order deliveries via the TIPTOP TABLES company fleet.



**FIGURE 1: LOADING BAY
STANDARD DIMENSIONS (IN MM)**

SUPPORTING FACILITIES

The design of the facility includes personnel facilities for office employees on site. The managing structure of one senior and two middle management have separate offices to allow for private consultations. The administrative and operational staff have different office spaces and layouts depending on the task-fulfilment.

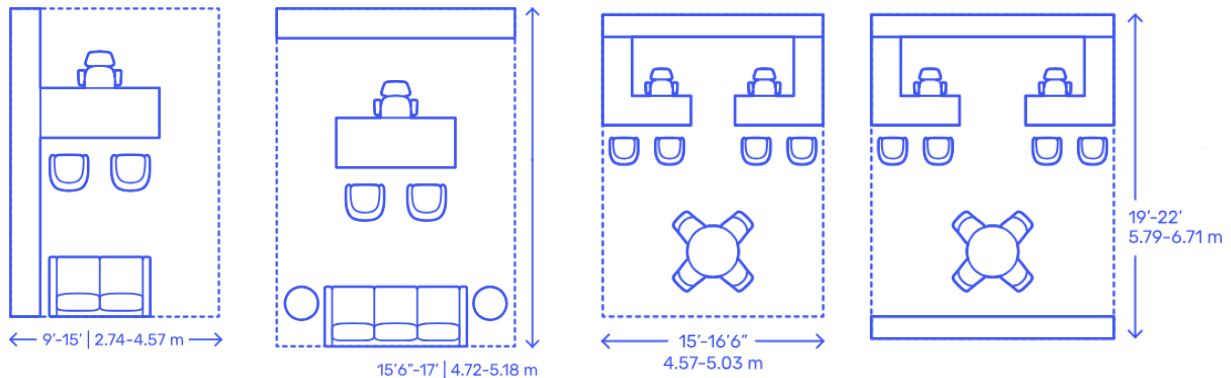


FIGURE 3: MANAGEMENT OFFICE LAYOUTS

FIGURE 2: STANDARD OFFICE LAYOUT

The facility has security considerations with a secure fence and security staff to monitor the entrance and exit of materials, vehicles, and people. The vehicles have designated parking for customers, employees, and company fleet.

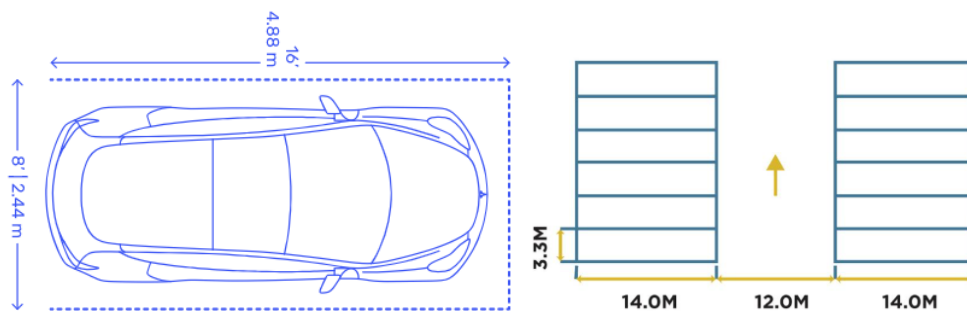


FIGURE 4: STANDARD VEHICLE AND TRUCK PARKING

The facility includes the following utilities:

- Kitchen for employee utilisation during lunch
- Sanitary facilities
- Water, plumbing, and ventilation requirements as regulated by the SABS
- Electricity and lighting infrastructures to power equipment and provide workstation visibility as employees do detailed work at workstations.
- Fire safety considerations

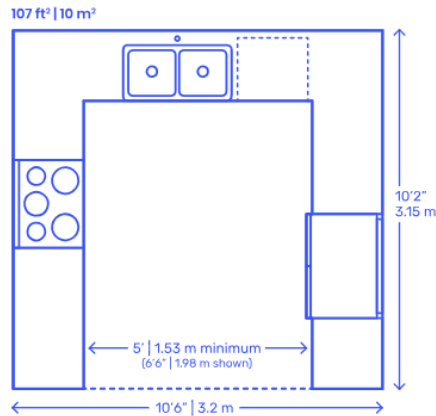


FIGURE 5: STANDARD KITCHEN LAYOUT

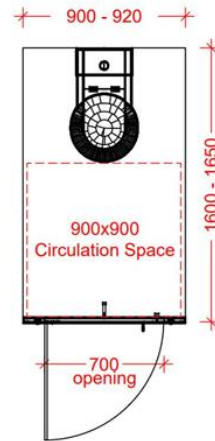


FIGURE 6: STANDARD BATHROOM STALL LAYOUT

4. Process Flow

The functionality of the facility is designed according to the production process, and the process flow diagram in Figure 7 was utilised to analyse these processes.

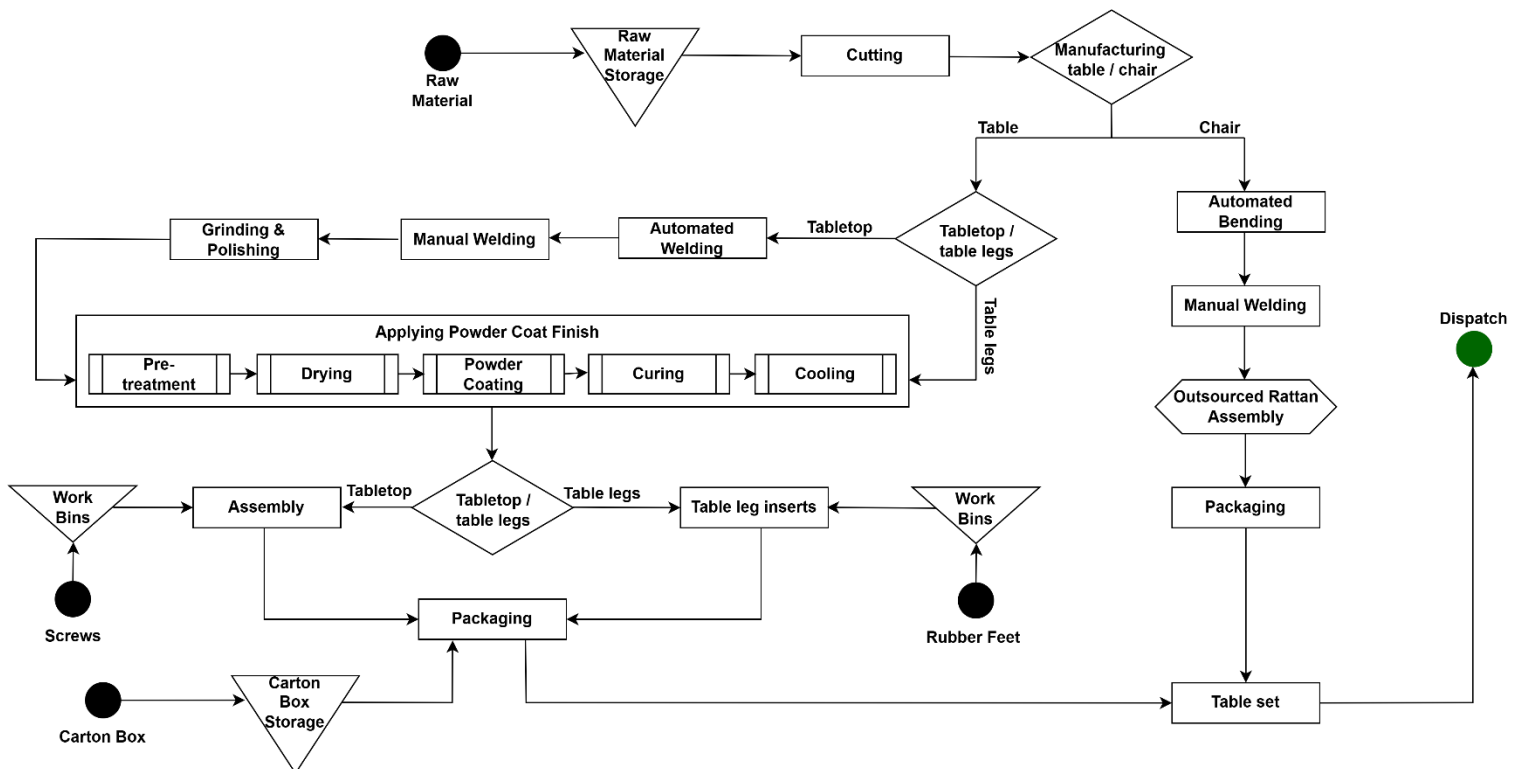


FIGURE 7: PROCESS FLOW DIAGRAM

5. Assumptions and Constraints

ASSUMPTIONS

- The expected consumer demand forecast is 5 000 per month.
- The company has an average of 20 working days per month.
- The company business week is 5 days per week, and a month has 4 weeks.
- The company operates on a 9-hour shift per day (08:00 – 18:00), and operators are given 30 minutes for lunch, and two 15-minute tea breaks per day.
- Overtime or a Saturday shift is only considered to provide flexibility in capacity.
- The workday is not affected by machine set-up time, machine changeover, and machine cleaning.
- The company expects a 20% growth rate in the near future, and the space utilisation calculations accounts for the increase.
- The space allocation allows for two weeks of raw material stock.
- The space capacity for finished goods stock storage is 4 weeks.
- A finished product set is composed of 5 tables and 20 chairs.
- A customer may collect a small order of less than 5 sets.
- Large orders are more than 5 sets and delivered via the company fleet.
- The standard aisle size is 1.5 m wide.
- Assume that raw material is delivered every 2 weeks.
- Assume that all the raw material is first cut to size by the CNC laser machine.
- The Rattan assembly onto the chair frame is outsourced, and 1000 chairs are delivered daily.
- Assume there are 2 supervisors, each oversees either table or chair manufacturing.

CONSTRAINTS

- The cutting process is subject to a 5% material scrap rate.
- The operators are restricted to move within the boundaries of the designated “operator aisle” as a safety precaution.
- There are workstations arrangement limitations due to movement safety and ventilation requirements.

Raw Materials:

The raw materials supplied by the supplier delivers the raw material in batches of 6m tubes and bars.

- Rectangular aluminium tubes (40mm x 20mm x 6000mm) to manufacture the tabletop frame, and slider.
- Aluminium flat bars (101,6mm x 9,52mm x 6000mm) forms the tabletop surface.
- Aluminium square tubes (60mm x 60mm x 6000mm) to manufacture the table legs.
- Aluminium round tubes (25,4mm x 1,6mm x 6000mm) to manufacture the chair frames.

Processing Times:

TABLE 1: PRODUCT PROCESSING TIME CONSTRAINTS

Process	Processing Time
Cutting of the table frames and chairs tubes	1 min per table & 1 min per chair
Automated welding of the slider structure	5 min per sliding structure
Manual welding of tabletop surface	10min per one tabletop frame
Manual grinding and polishing operations	1min per tabletop
Table structures powder coating	30 sec per structure
Manual tabletops and slider assembly	5 min per table
Inserting rubber feet onto table legs	1min per 4 legs
Table set packaging	5min per set.
Chair frame bending	3min per chair
Manual welding of chair frame	5 min per chair
Chair stacking	10min to stack 20 chairs

6. Detail Design Specifications

DEMAND

The raw material requirement was calculated to meet the monthly customer demand of 5000 sets. The chairs production is four times that of tables (monthly demand = 20000), as a set includes one table and four chairs. The daily, and hourly production requirements are calculated in Equation 1 to 4, and summarised in Table 2.

$$\begin{aligned}
 \text{daily set production} &= \frac{\text{monthly forecast}}{\text{working days per month}} \\
 &= \frac{5000 \text{ sets}}{20 \text{ working days per month}} \\
 &= 250 \text{ sets per day}
 \end{aligned} \tag{1}$$

$$\begin{aligned}
 \text{hourly set production} &= \frac{\text{daily set production}}{8 \text{ hour work day}} \\
 &= \frac{250 \text{ sets}}{8 \text{ hour work day}} \\
 &= 31.25 \approx 32 \text{ sets per hour}
 \end{aligned} \tag{2}$$

$$\text{daily chair production} = \frac{\text{monthly forecast}}{\text{working days per month}} \times 4 \tag{3}$$

$$\text{hourly chair production} = \frac{\text{daily chair production}}{8 \text{ hour work day}} \tag{4}$$

TABLE 2: PRODUCTION DEMAND REQUIREMENTS

Product	Monthly Production (units/month)	Daily Production (units/day)	Hourly production (units/hour)
Table	5000	250	32
Chair	20000	1000	125

6.1. Raw Material Requirements

There is a 2 week (10 days) raw material stock storage stock requirement. The following raw material assumptions were derived from the product design specifications depicted in Figure 8 – 10:

- A tabletop (table frame) structure is manufactured from one 6m rectangular tube, and one table requires 2 tabletops.
- A slider structure is manufactured from one 6m rectangular tube.
- A tabletop surface is manufactured from 2.5 flat bars.
- One 6m square tube can manufacture 8 table legs (two tables).

$$6m \text{ square tube} \div 0.7 \text{ leg length} = 8.57 \approx 8 \text{ table legs}$$
- A chair frame structure is manufactured from one 6m round tube.

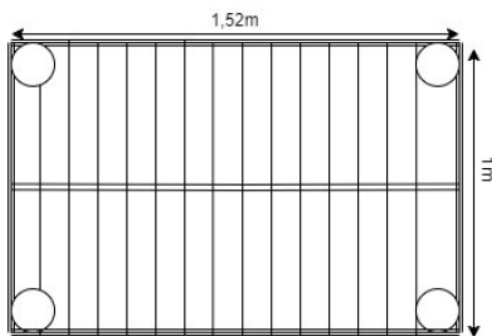


FIGURE 8: TABLETOP TOP VIEW

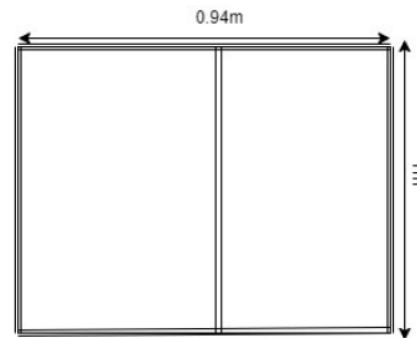


FIGURE 9: SLIDER STRUCTURE TOP VIEW

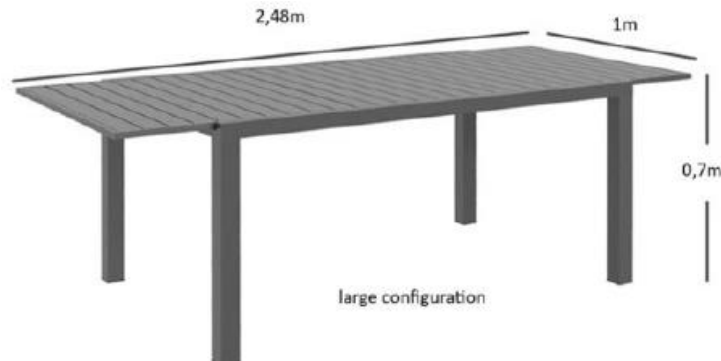


FIGURE 10: LARGE TABLE CONFIGURATION

Equations 5 to 8 are used to determine the raw material required to fulfil the two-week raw material inventory requirements. The cutting process is the first manufacturing step, and all the raw material is thus subject to a 5% scrap rate that occurs during this process, calculated via Equation 9 and the raw material requirement is summarised in Table 3 (rounded up to the nearest whole number).

$$\text{rectangular tubes required} = \text{daily sets} \times (2 \text{ tableframes} + 1 \text{ slider}) \times 10 \text{ days} \quad (5)$$

$$\text{flat bars required} = 2.5 \text{ flat bars} \times \text{daily sets} (2 \text{ tableframes}) \times 10 \text{ days} \quad (6)$$

$$\text{square tubes required} = \text{daily sets} \div 2 \text{ tables per tube} \times 10 \text{ days} \quad (7)$$

$$\text{round tubes required} = \text{daily chair production} \times 10 \text{ days} \quad (8)$$

$$\text{raw material adjusted for scrap} = \text{raw material product} \times 1.05 \quad (9)$$

TABLE 3: RAW MATERIAL INVENTORY REQUIREMENT (IN RAW MATERIAL UNITS)

Raw Material Product	Daily Inventory Stock	Inventory Stock	Inventory Stock Adjusted for Scrap
Rectangular tubes	750	7500	7875
Flat bars	1250	12500	13125
Square tubes	125	1250	1313
Round tubes	1000	10000	10500

The raw material is delivered in bundles of 20 (Figure 11) as summarised in Table 4 and delivered per crate load, separated per raw material product. The crate size is 6000mm × 1000mm × 1500mm depicted in Figure 12, and the raw material is packed according to the configuration in Figure 13.

$$\text{raw material bundles requirement} = \text{raw material adjusted for scrap} \div 20 \quad (10)$$

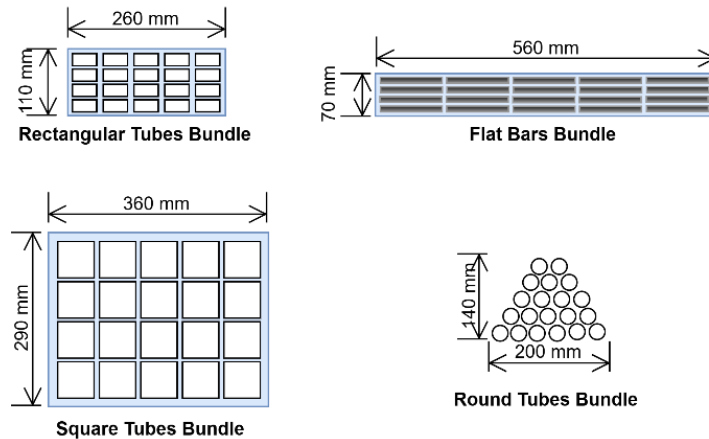


FIGURE 11: RAW MATERIAL BUNDLES DIAGRAM



FIGURE 12: RAW MATERIAL CRATE

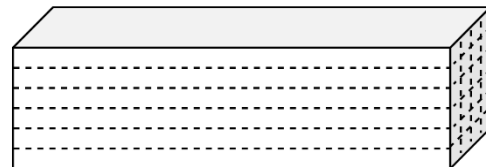


FIGURE 13: RAW MATERIAL CRATE CONFIGURATION

Material Handling

The number of pallets delivered every 2 weeks is determined by Equation 11 and summarised in Table 4. The raw material is offloaded via a forklift, Figure 14, and stacked onto the racks in Figure 15.

$$\text{biweekly pallet delivery} = \text{raw material bundles} \div (\text{rows} \times \text{columns}) \quad (11)$$

TABLE 4: RAW MATERIAL DELIVERIES (IN UNITS)

Raw Material Product	Rows	Columns	Bundle Requirement	Biweekly Delivery (pallets every 2 weeks)
Rectangular tubes	3	13	394	11
Flat bars	2	21	657	16
Square tubes	3	4	66	6
Round tubes	7	7	525	11
Total biweekly pallet delivery				44



FIGURE 14: ELECTRIC FORKLIFT TRUCK

TABLE 5: FORKLIFT SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	3.33 m x 1.2 m x 2 m
Load Capacity	2.5 t
Operator	1
Machine cost	R 175 212.00

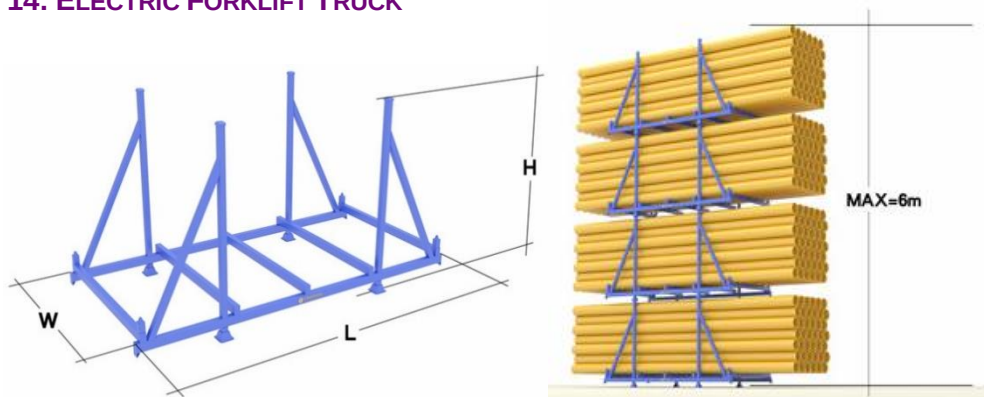


FIGURE 15: PALLET STORAGE RACKS

TABLE 6: PALLET STORAGE RACK SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	2.3 m x 1.1 m x 1.3 m
Load Capacity	2.5 t
Equipment cost	R 3550,95

A total of 36 pallets are delivered biweekly, as such 36 storage racks are required, of which 4 can be stacked on top of each other. The total floor space requirement is calculated in Equation 12.

$$\begin{aligned}
 \text{total stacked racks} &= 44 \div 4 = 11 \text{ stacked racks} \\
 \text{Raw material storage area} &= (\text{pallet } L \times \text{rack } W \times 9) + 20\% \\
 &= (6m \times 1.1m \times 11) + 20\% \\
 &= 87.12 \text{ m}^2
 \end{aligned}
 \tag{12}$$

The raw material is unboxed to satisfy the daily production requirements (Equation 13) and stacked onto a portable metal storage trolley, Figure 16. The portable trolleys facilitate transporting the raw material to the cutting process, each trolley contains one raw material product and therefore 4 trolleys are required.

$$\text{trolley capacity} = \text{daily inventory stock} \div 20
 \tag{13}$$

TABLE 7: TROLLEY CAPACITY

Raw Material Product	Trolley Capacity (bundle units)
Rectangular tubes	38
Flat bars	63
Square tubes	7
Round tubes	50



FIGURE 16: PORTABLE METAL STORAGE TROLLEY

TABLE 8: PORTABLE METAL TROLLEY SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	1.5 m x 1.2 m x 1.2 m
Load Capacity	500 kg
Operator	1
Equipment cost	R 2700,00

$$\begin{aligned}
 \text{Portable trolley storage area} &= (\text{trolley } L \times \text{trolley } W \times 4) + 20\% \\
 &= (1.5m \times 1.2m \times 4) + 20\% \\
 &= 8.64 \text{ m}^2
 \end{aligned}
 \tag{14}$$

The aisle space required for the forklift accommodates the length of the pallet crate and enable turning, similarly the aisle size of the portable trolley rack accommodates the 6m length of the raw material.

$$\text{forklift aisle width} = \text{turning radius} + \text{fork length} + \text{load length} + \text{safety clearance} \quad (15)$$

$$= (2 + 1.07 + 6 + 0.2) + 20\% = 11.12 \text{ m}$$

$$\text{trolley aisle width} = \text{turning radius} + \text{load length} + \text{safety clearance} \quad (16)$$

$$= (1.5 + 6 + 0.2) + 20\% = 9.24 \text{ m}$$

The receiving facility layout is discussed in section 6.4 (Receiving & Dispatch).

6.2. Manufacturing Time

The manufacturing processes have processing time constraints as mentioned in Table 1, and with the daily production information from Table 2 the machine requirements were calculated via Equation 17 and summarised in Table 9. Note that available processing time is $8 \text{ hr work day} \times 60 \text{ mins} = 480 \text{ min/day}$.

Assumptions

- The table is composed of 4 parts which are then powder coated, namely 2 tabletops, 1 slider, and 1 legs (the four leg components are treated as a unit)
- The tables and chairs are processed by the same CNC cutting machine, and the same manual welding equipment is used for processing the tables and chairs.
- The machines and operators work at an 80% efficiency rate.

$$\text{number of machine or operators required} = \frac{\text{processing time}}{\text{available processing time}} \quad (17)$$

TABLE 9: MACHINE AND OPERATOR REQUIREMENTS

Process	Daily Production (units/day)	Processing Time (processing time × units processed)	Number of Machines or Operators Required
Cutting of table & chair raw material	263 tables 1050 chairs	1313 mins	2.74 ≈ 3 machines
Automated sliding structure welding	250 sliders	1250 mins	2.60 ≈ 3 machines
Manual tabletop & chair welding	500 tabletops 1000 chairs	10000 mins	20.83 ≈ 21 operators
Tabletop grinding and finishing	500 tabletops	500 mins	1.04 ≈ 2 operators
Powder coating table structures	1000 table structures	500 mins	1.04 ≈ 2 machines
Table assembly	250 tables	1250 mins	2.60 ≈ 3 operators
Applying rubber inserts to legs	250 tables	250 mins	0.52 ≈ 1 operator
Table packaging	50 sets	250 mins	0.52 ≈ 1 operator
Bending chair frame	1000 chairs	3000 mins	6.25 ≈ 7 machines
Stacking chairs	50 stacks	500 mins	1.04 ≈ 2 operators

6.3. Manufacturing Processes

CUTTING

Process

A computer numerical control (CNC) laser cutting machine is used to cut materials at a high precision rate to reduce material waste during the manufacturing process. The laser cutting machine focuses a high-energy beam to trigger rapid melting of the target material and enables the cutting of materials of various size, including round pipes, square tubes, rectangular tubes, and flat bars. The cut piece is then released from the machine, ready to be removed by an operator.

Machine



TABLE 10: CNC LASER MACHINE SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	13.5 m x 3.2 m x 2.5 m
Load Material	Metal
Operator	1
Machine cost	R 530 500,20

FIGURE 17: CNC LASER MACHINE

Cutting Workstation

The CNC laser machine is automatic; however, it requires an operator to feed the raw material into the machine and collect the WIP parts from the machine. An operator aisle is provided to enable the operator to safely move between the raw material loading area, computer interface, and the WIP parts collection. The total floor space of the laser cutting workstations was calculated using Equation 18.

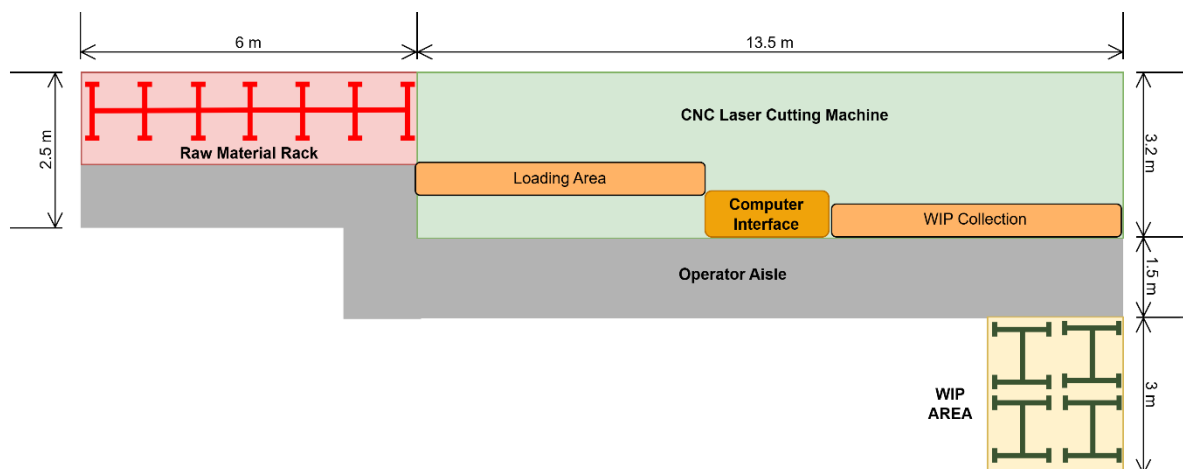


FIGURE 18: LASER CUTTING WORKSTATION

$$\begin{aligned} \text{total floor space} &= \text{number of stations} \left(\frac{\text{machine area} + \text{raw material area}}{\text{+WIP area} + \text{aisle space}} \right) + 20\% \quad (18) \\ &= 3(43.2 + 6 + 6.9 + 29.25) + 20\% = 307.26 \text{ m}^2 \end{aligned}$$

The waste management during the cutting process is via a metal scrap bin, the scrap bin can accommodate 6t of metal waste which is sufficient to handle a week's worth of aluminium scraps until it is collected and discarded by a third-party collector.



FIGURE 19: METAL SCRAP BIN

TABLE 11: METAL SCRAP BIN SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	2.5 m x 1.3 m x 1.3 m
Load Material	Aluminium Scraps
Operator	1
Equipment Cost	R 10 400.00

Material Handling

The WIP parts are loaded onto a tube cart to move the long materials, the tiers allow separation of the varying tubes and bars, and the stop at the end prevents objects from falling off. As seen in Figure 20, 4 carts are utilised, and a cart is stacked on an hourly basis after which it is transported to the next process.



FIGURE 20: TUBE CART

TABLE 12: TUBE CART SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	1.3 m x 1.1 m x 1.8 m
Load Material	Aluminium tubes & bars
Operator	1
Equipment cost	R 1700,00

AUTOMATIC WELDING

Process

The automatic laser welding robot is a 6-axis robot with high flexibility, which can complete the welding of complex workpieces. The workpiece is placed onto the welding stand by an operator, the operator then starts the welding machine, and when the welded piece is complete, the operator removes it from the welding stand.

Machine



FIGURE 21: AUTOMATIC LASER WELDING ROBOT

TABLE 13: AUTOMATIC WELDING ROBOT SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	1.3 m x 1.1 m x 1.8 m
Load Material	Aluminium rectangular tubes
Operator	1
Machine cost	R190 300,50

Automatic Welding Workstation

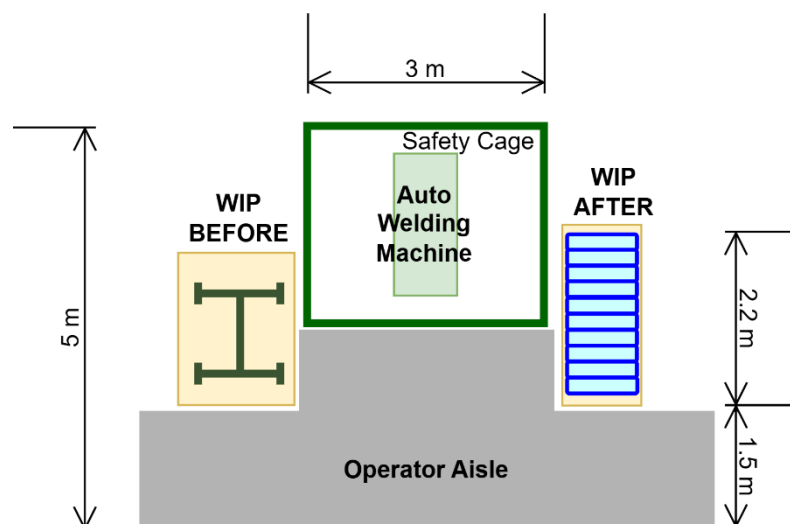


FIGURE 22: AUTOMATIC WELDING MACHINE

The workstation in Figure 22 includes a 3m x 3m Safety Cage where the robot is housed to ensure the safety of the operator, as well as a designated operator aisle. The total floor space was calculated using Equation 19.



FIGURE 23: AUTOMATIC WELDING MACHINE SAFETY CAGE

$$\begin{aligned}
 \text{total floor space} &= \text{number of stations} \left(\begin{array}{l} \text{machine area} + \text{WIP before} \\ + \text{WIP after} + \text{aisle space} \end{array} \right) + 20\% \quad (19) \\
 &= 3(9 + 2.86 + 2.86 + 14.1) = 103.75 \text{ m}^2
 \end{aligned}$$

Material Handling

The WIP pieces are delivered via the tube cart from the cutting process and then placed onto the robot weld-stand. After the completion of the WIP part it is loaded onto the remnant cart, where it is transported to the next process.



FIGURE 24: REMNANT CART

TABLE 14: REMNANT CART SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	2.2 m x 0.8 m x 1.2 m
Operator	1
Equipment cost	R 3800,00

MANUAL WELDING

Process

The manual welding process involves an operator joining the aluminium flat bars to the table frame to form the tabletops of the table, and making the chair frame. The operator receives the parts from the automatic welding station.

Machine



TABLE 15: MANUAL WELDING EQUIPMENT

Equipment Specifications	
Dimensions (L x W x H)	0.9 m x 0.6 m x 1.8 m
Load Material	Table and chair frames
Operator	1
Machine cost	R 35 879.00

FIGURE 25: MANUAL WELDING EQUIPMENT

Manual Welding Workstation

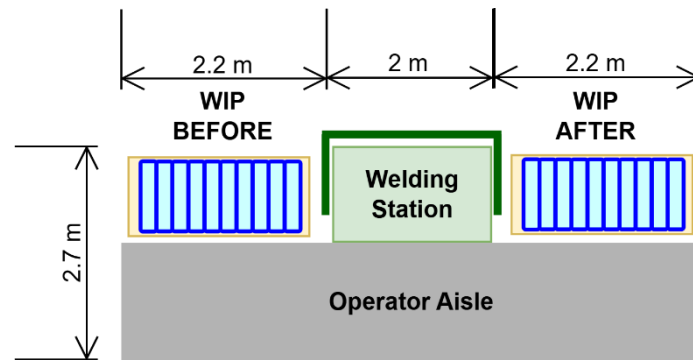


FIGURE 26: MANUAL WELDING WORKSTATION

The workstation in Figure 26 includes a 2m x 1m safety boundary ensures safety by protecting other operators passing by from sparks, intense light, and fumes. It also defines the work area which prevents unauthorized access and reducing the risk of accidents. Additionally, the welding and grinding stand base (2m x 1.2m x 0.9 m) is equipped with holes for the extraction of dust and gases generated during grinding or welding.

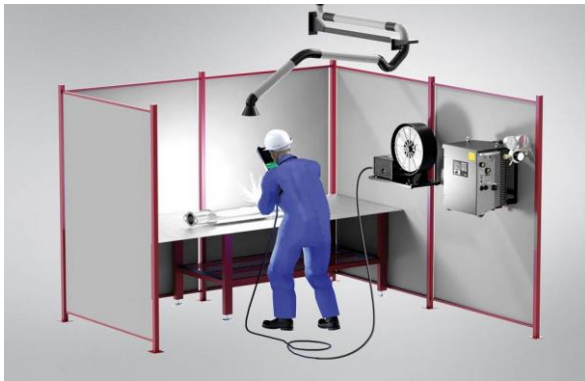


FIGURE 27: MANUAL WELDING SAFETY BOUNDARY



FIGURE 28: WELDING AND GRINDING STAND BASE

The total floor space was calculated by using Equation 19:

$$total\ floor\ space = 21(2.4 + 1.76 + 1.76 + 9.6) + 20\% = 391.10\ m^2$$

Material Handling

The WIP pieces are delivered via the remnant cart and transported to the next process with a remnant cart, the remnant cart is depicted in Figure 24. (The material handling for all the processes is similar, until after assembly workstation)

GRINDING AND POLISHING

Process

The tabletop structures obtained from the welding processes are grinded and polished manually to achieve a smooth finish.

Machine



TABLE 16: ANGLE GRINDER SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	0.3 m x 0.3 m x 0.1 m
Load Material	Tabletop structures
Operator	1
Machine cost	R 1 900.00

FIGURE 29: ANGLE GRINDER

Grinding and Polishing Workstation

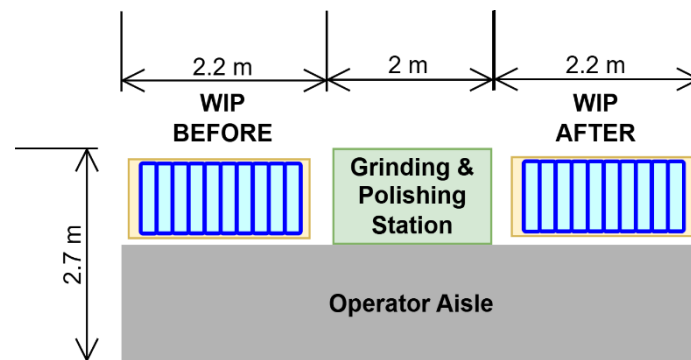


FIGURE 30: GRINDING AND POLISHING WORKSTATION

The grinding station utilises the same 2m x 1.2m x 0.9 m welding and grinding stand base depicted in Figure 28, to extract the dust and particles generated during the grinding process. An operator aisle is a safety precaution, similar to the aforementioned processes.

The total floor space was calculated by using Equation 19:

$$total\ floor\ space = 2(2 + 1.76 + 1.76 + 9.6) + 20\% = 36.29\ m^2$$

POWDER COATING

Process

The powder coating process contains sub-processes explained below [1]:

- Pre-treatment process cleans and prepares the aluminium surfaces for powder coating to promote adhesion and finish quality. The aluminium parts are cleaned through a chemical rinse, manually applied at the wash station.
- The pre-treated parts are then fed through drying ovens to dry the parts.

- The parts are then ready to be fed through the automated powder coating booths; the booths contain a ventilation system which ensures proper airflow to capture overspray and maintain a clean environment.
- The powder coat is melted and fused onto the surface of the parts during the curing process as the parts are fed through the curing oven.
- The cured parts are fed through a cooling chamber to cooldown gradually to avoid coat warping or defects.

Machines



FIGURE 31: METAL RINSING EQUIPMENT

TABLE 17: METAL RINSING EQUIPMENT SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	0.1 m x 0.05 m x 1 m
Load Material	Table structures
Operator	1
Machine cost	R 5 300,50



FIGURE 32: DRYING AND CURING OVEN

TABLE 18: OVEN SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	5 m x 0.9 m x 2 m
Load Material	Table structures
Machine cost	R 1 595 000,00



FIGURE 33: POWDER COATING EQUIPMENT

TABLE 19: POWDER COATING SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	5 m x 0.8 m x 2 m
Load Material	Table structures
Machine cost	R 580 000,00

Powder Coating Workstation

The workstation for the powder coating process is depicted in Figure 34. As this workstation has multiple sub-process, the total floor pace was calculated as the workstation area.

total floor space

$$= nr \text{ of stations}(L \times B) + 20\%$$

$$= 2(35 \text{ m} \times 13 \text{ m}) + 20\% = 1092 \text{ m}^2$$

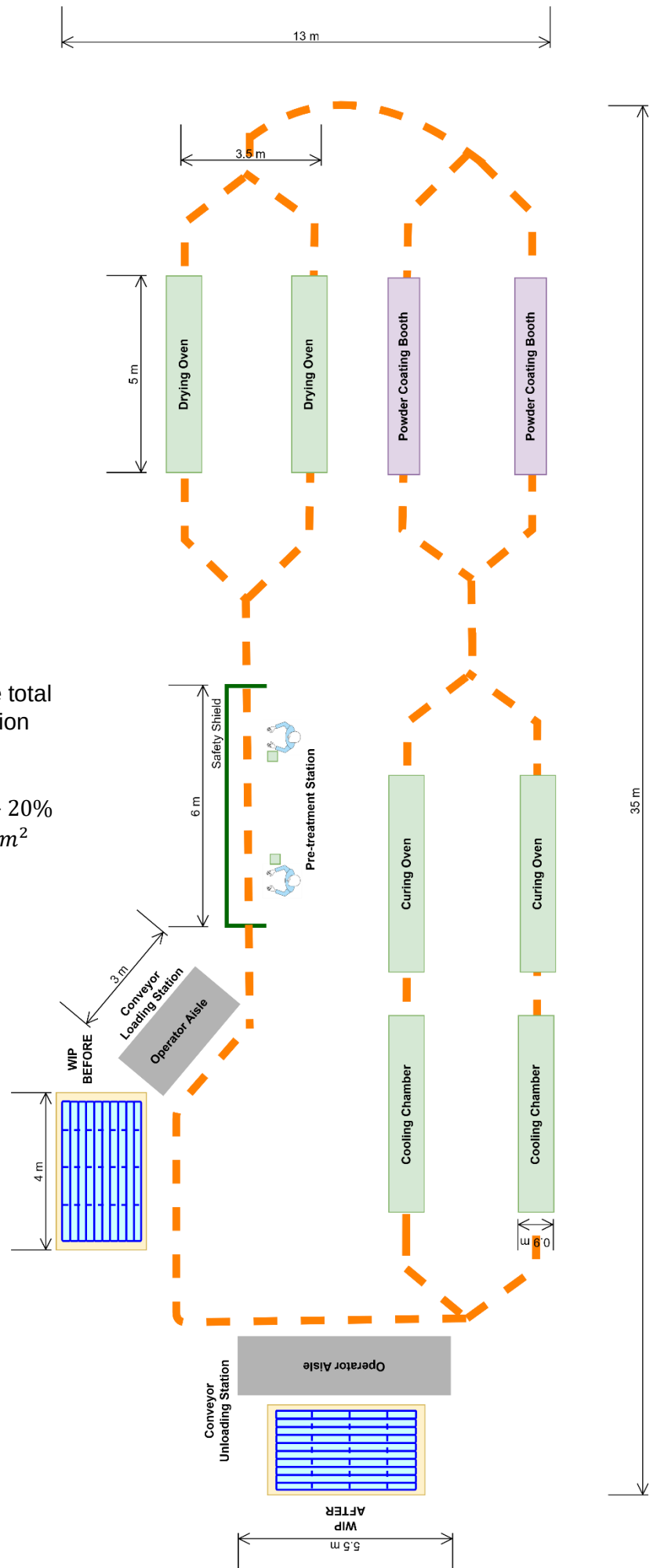


FIGURE 34: POWDER COATING PROCESS WORKSTATION

Material Handling

The powder coating process utilises an overhead conveyor system that hangs parts from hooks for continuous production with loading and unloading stations to move parts onto and off the conveyor system. A technician (operator) ensures that the conveyor speed and break time happens efficiently.



FIGURE 35: OVERHEAD CONVEYOR SYSTEM

TABLE 20: CONVEYOR SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	95 m x 2.3 m x 2 m
Load Material	Table structures
Operator	1
Machine cost	R 1 775 000,00

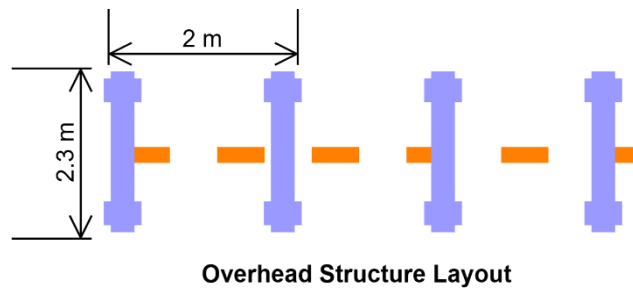


FIGURE 36: CONVEYOR ASSEMBLY DIAGRAM

The pillars are not indicated on the workstation diagram, but Figure 36 illustrates the overhead conveyor system structural assembly. The 95 m length dimension, it the overall length of the conveyor system.

TABLE ASSEMBLY

Process

The table assembly involves manually joining two tabletops with the sliding structure to create the 2-in-1 table. The operators align the parts, fit the sliding mechanism, and secure components using a screwdriver/drill to screw in 20 screws per table. The process is made more convenient as the equipment is battery operated, allowing for more operator flexibility.

Machine



FIGURE 37: SCREWDRIVER/DRILL EQUIPMENT

TABLE 21: SCREWDRIVER/DRILL SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	0.2 m x 0.3 m x 0.1 m
Load Material	Tabletop structures
Operator	1
Machine cost	R 1 250.00

Assembly Workstation

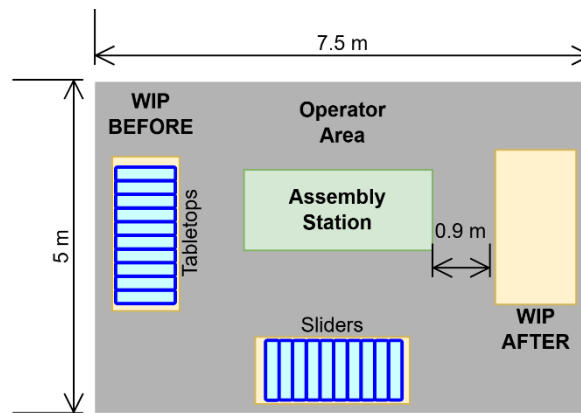


FIGURE 38: ASSEMBLY WORKSTATION

The workstation has a central flow element, where the WIP before and WIP after are on either side of the table being assembled. This layout allows for minimal movement between areas, while allowing enough space to enable comfortable motion.

The total area was calculated using the length and width dimensions of the workspace:

$$\begin{aligned}
 \text{total floor space} &= \text{number of stations}(L \times B) + 20\% \\
 &= 3(7.5\text{m} \times 5\text{m}) + 20\% = 135\text{m}^2
 \end{aligned}
 \tag{20}$$

The assembly work bench includes overhead lights, screw storage bins, and shall be equipped with cabinets to house the screwdriver/drill equipment, as well as electric wall outlets to plug in the drill to charge when not in use.



FIGURE 39: ASSEMBLY WORK BENCH

TABLE 22: ASSEMBLY WORK BENCH SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	2.8 m x 1.2 m x 0.9 m
Load Material	Tabletop structures
Operator	1
Equipment cost	R 10 400,50

Material Handling

The WIP before parts are received via the remnant carts, and since the cart has 10 slots the parts received can assemble 2 table sets of 5 (10 tables), which fulfils the hourly production criteria. The WIP after is loaded onto the disassembled wire mesh trolley cage, and once all the sets are completed the trolley is assembled and ready to be transported to the set packing process. The wire mesh cage is lined with a soft material to prevent scratches to the table part surfaces.



FIGURE 40: METAL TROLLEY CART

TABLE 23: METAL TROLLEY CART SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	2.3 m x 1.2 m x 1 m
Operator	1
Equipment cost	R 1 400,00

RUBBER LEVELERS ASSEMBLY

Process

The process of inserting rubber feet onto the table legs involves manually pressing the rubber feet into place. This feature makes the table more stable and prevents wobbling. The work bench in Figure 41 can be tilted to make assembly easier, the rubber feet are contained in the bins above, and an overhead light assists with visibility. The rubber feet are 60 mm x 60 mm and supplied by the online marketplace, eBay [2].

Equipment



FIGURE 41: RUBBER ASSEMBLY WORK BENCH

TABLE 24: RUBBER WORK BENCH SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	1.2 m x 1.5 m x 0.9 m
Load Material	Tablet legs
Operator	1
Equipment cost	R 10 400,50

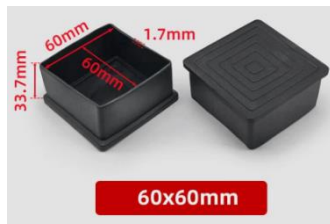
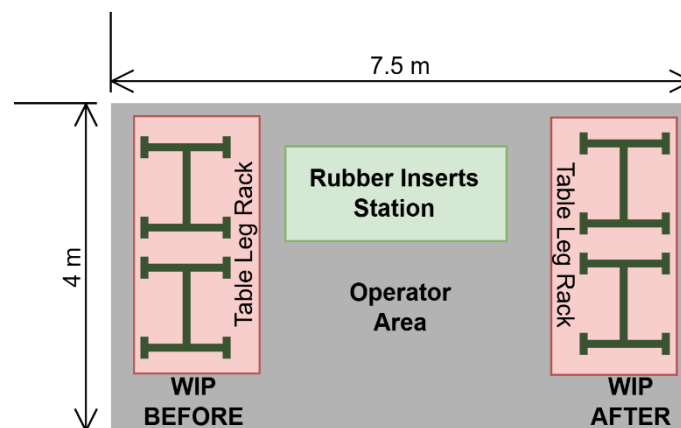


FIGURE 42: RUBBER FEET INSERTS

TABLE 25: RUBBER FEET INSERTS SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	60mm x 60mm x 33.7mm
Load Material	Table legs
Material cost	R 1360,00 / 50 pieces

Rubber Assembly Workstation



Similar to the table assembly workstation, the material has been placed on either side of the work bench which allows for minimal movement. The total area was calculated using Equation 20:

$$total\ floor\ space = 1(7.5m \times 4m) + 20\% = 36m^2$$

Material Handling

Tube carts, depicted in Figure 20, is used to facilitate WIP before and WIP after material handling.

TABLE SETS PACKAGING

Process

The table sets are hand packaged which serves as a final inspection to ensure each table meets quality standards. A packaged set consists of 5 tables, along with their legs, are packed into one carton box.

Equipment

An assembly work bench, Figure 39, is utilised to group the tabletops and table legs for a more efficient packaging process. The work bench is modified and does not include the stands for work bins and overhead lighting.

Table Set Packaging Workstations

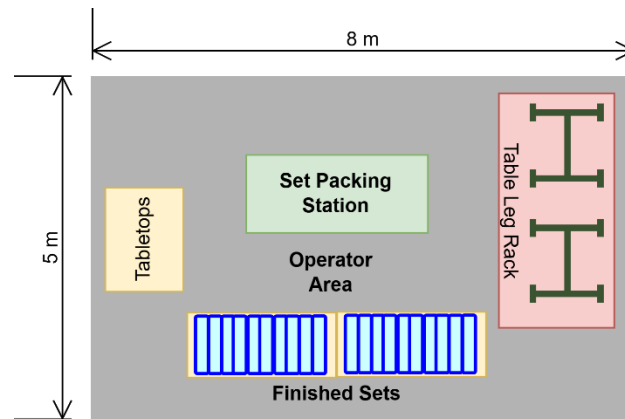


FIGURE 43: TABLE SET GROUPING WORKSTATION

The material has been placed on either side of the work bench which allows for minimal movement. The total area was calculated using Equation 20:

$$\text{total floor space} = 1(8m \times 5m) + 20\% = 48 m^2$$

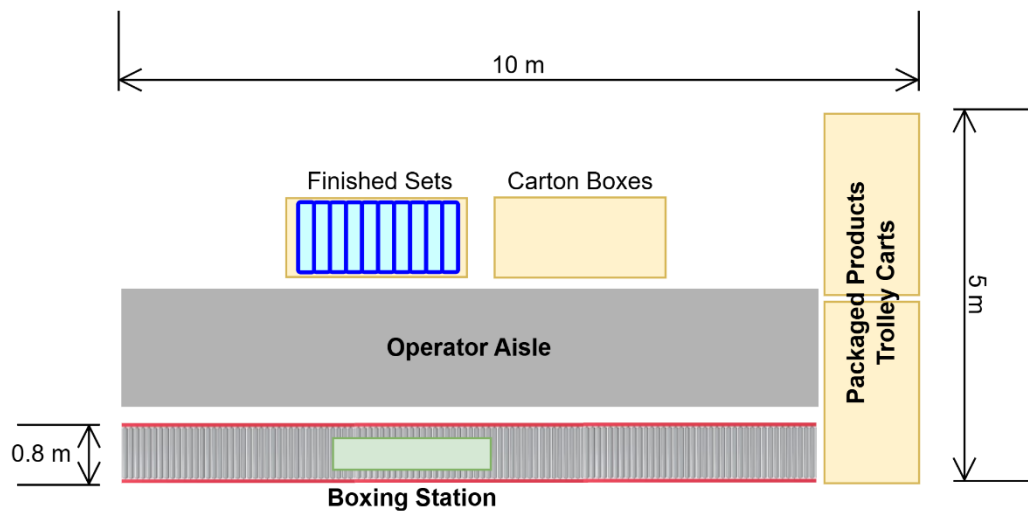


FIGURE 44: TABLE SET BOXING WORKSTATION

The finished sets are transported to the boxing workstation, where 5 sets are placed into carton boxes. The carton boxes are in close proximity to the boxing station. The total area was calculated using Equation 20:

$$\text{total floor space} = 1(10m \times 5m) + 20\% = 60 m^2$$

Material Handling

The tabletops are transported using the metal trolley cart, the table legs are transported using the tube cart, and the finished sets are transported using a remnant cart. The empty carton box is transported via metal trolley carts. The empty carton box is placed on a floor roller conveyor, the carton box is packaged, and the sealed box is moved on a roller conveyor to be stacked on a metal trolley cart (Figure 40).

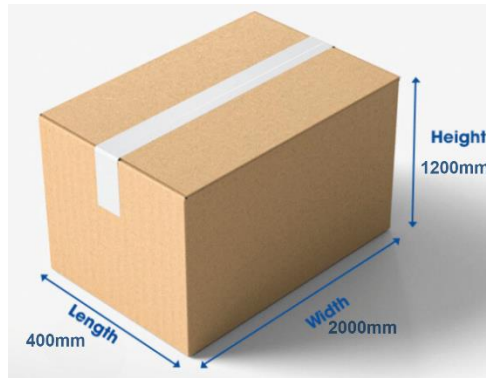


FIGURE 45: CARTON BOX DIMENSIONS



FIGURE 46: ROLLER CONVEYOR

TABLE 26: ROLLER CONVEYOR SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W)	9m x 0.5m
Equipment cost	R 45 000.00

AUTOMATED BENDING

Process

The automated bending of tubes to the required angles and shapes forms the frame parts of the chair at high accuracy and uniformity.

Equipment



FIGURE 47: AUTOMATED BENDING MACHINE

TABLE 27: BENDING MACHINE SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	3.9 m x 0.7 m x 1.2 m
Load Material	Chair tubes
Operator	1
Machine cost	R 89000,00

Automated Bending Workstation

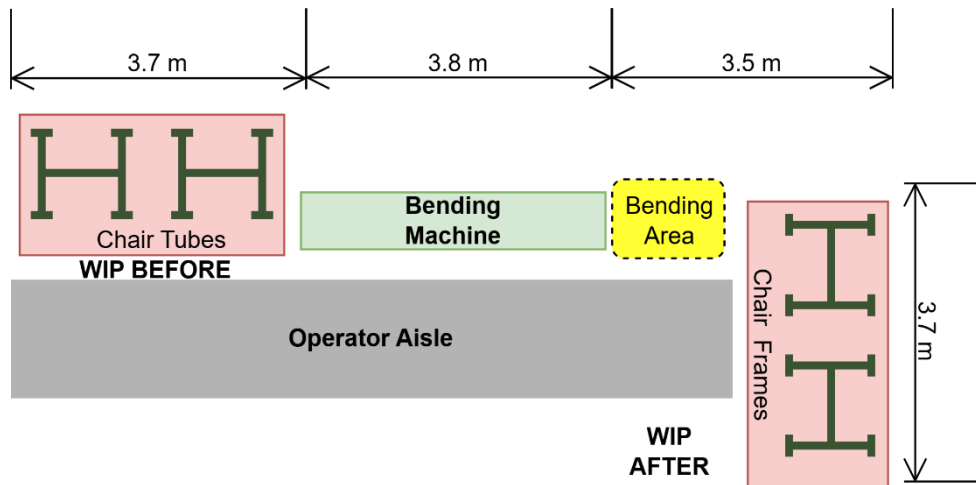


FIGURE 48: AUTOMATED BENDING WORKSTATION

The operator feeds the into the bending machine, the operator then places the finished WIP part onto a trolley rack. The total floor space was calculated by using Equation 19:

$$\text{total floor space} = 7(2.73 + 7.4 + 7.4 + 13.5) + 20\% = 260.65 \text{ m}^2$$

RATTAN OUTSOURCING

Process

The process of attaching rattan to the chair frames is outsourced to specialized vendors, where skilled workers manually weave or attach synthetic rattan onto the frames.

Material Handling

The rattan chairs are delivered in a similar fashion as the raw material tubes and bars. The rattan chairs are delivered per crate load where the crate size is $6000\text{mm} \times 1000\text{mm} \times 700\text{mm}$ depicted in Figure 12, and the chairs are packaged to the configuration in Figure 50, 2 rows of 25 chairs are delivered this way. Therefore, 50 chairs arrive per pallet crate, and thus 20 pallet crates are delivered every day.



FIGURE 49: RATTAN DELIVERY CONFIGURATION

The pallets are stacked similarly to Figure 15, and the total floor space requirement is calculated via Equation 12:

$$\text{total stacked racks} = 20 \div 4 = 5 \text{ stacked racks}$$

$$\text{Rattan chair storage area} = (6\text{m} \times 0.7\text{m} \times 5) + 20\% = 25.2 \text{ m}^2$$

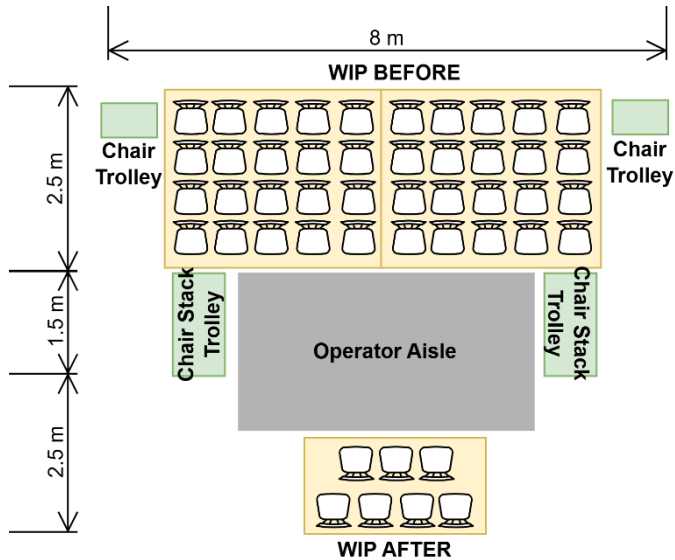
The outsourced rattan chairs receiving area is discussed in section 6.4 (Receiving & Dispatch).

CHAIR STACKING

Process

Chairs are stacked and secured with tie-straps, awaiting combination with tabletops to form complete sets.

Chair Stacking Workstation



The chairs are unloaded from the pallet crate as received from the rattan processing supplier. The chairs are unloaded to facilitate the hourly production requirements and placed onto the WIP before area; 5 chairs are stacked onto each other. The chairs are then stacked onto a chair stack trolley, tied together with tie-straps, and placed on the WIP after area. The total area was calculated using Equation 20:

$$\begin{aligned} \text{total floor space} &= 1(8\text{m} \times 6.5\text{m}) + 20\% \\ &= 62.40 \text{ m}^2 \end{aligned}$$

FIGURE 50: CHAIR STACKING WORKSTATION

Material Handling



FIGURE 51: CHAIR TROLLEY

TABLE 28: CHAIR TROLLEY SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	1.1m x 0.5m x 0.8m
Operator	1
Equipment cost	R 2 200,50

The chairs are unloaded from the supplier pallet crate and transported to the WIP before area via chair trolley.

The chair stacking trolley has been customised to facilitate stacking 20 chairs. To accommodate operator height requirements, a step ladder is utilised.



TABLE 29: CHAIR STACKING TROLLEY SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	1.5 m x 0.8 m x 2.4 m
Operator	1
Equipment cost	R 9 800,75

FIGURE 52: CHAIR STACKING TROLLEY

6.4. Receiving & Dispatch

Process

The raw materials and rattan chair WIP components are received via the loading bay and stored in the designated goods receiving areas. The finished goods stock requires a storage capacity to retain 4 weeks' worth of products to be exported overseas via shipments. Additionally, the distribution storage area facilitates customer collections (small orders), and large order truck dispatch. *Refer to Appendix A for enlarged images of the storage area layouts.*

Raw Material Receiving Storage Area

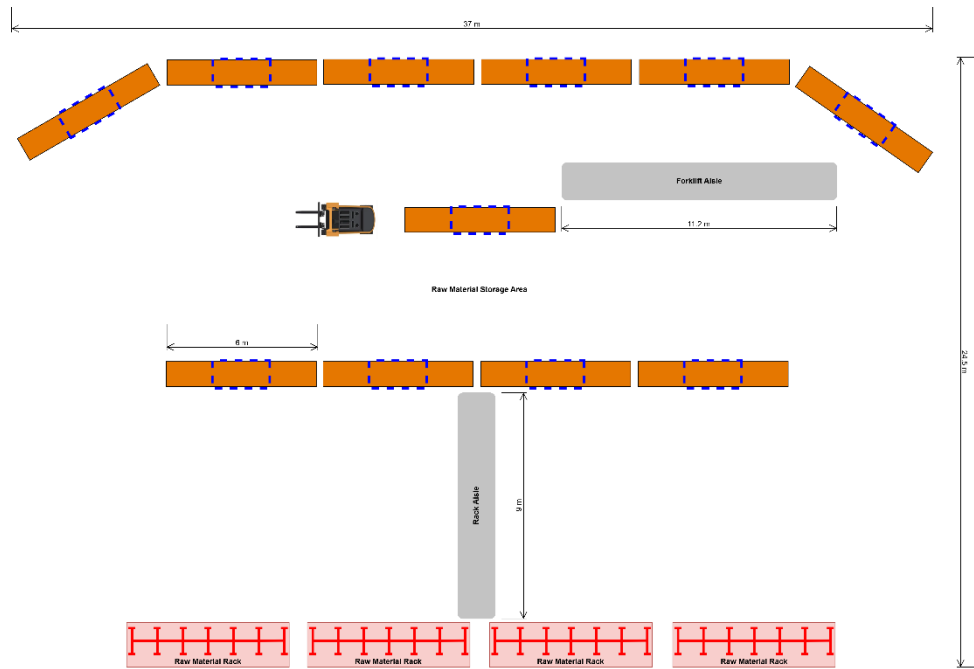


FIGURE 53: RAW MATERIAL STORAGE LAYOUT

Rattan Product Storage Layout

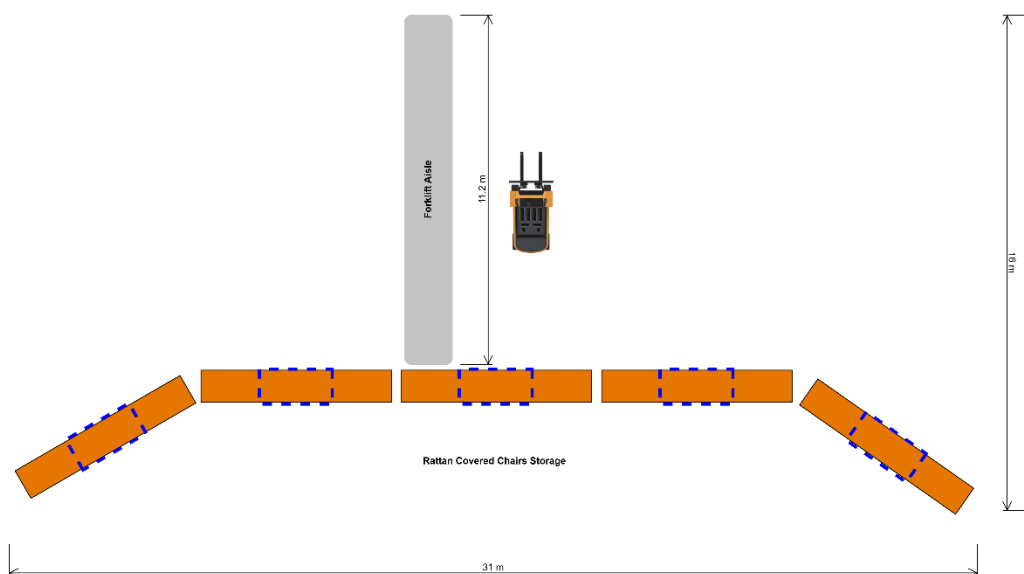


FIGURE 54: RATTAN PRODUCTS STORAGE LAYOUT

Finished Goods Storage Layout

- Assume the facility already has a 4-week shipment stock on hand

The packaged boxes (table sets) are stacked onto metal trolley carts as in Figure 40, the boxes are transported to and packed into containers, Figure 56. The chair stacks are transported as units, secured by the tie-straps, and also packed into their respective containers. The containers are loaded onto shipment trucks via the forklift in Figure 14.



FIGURE 55: CONTAINER DIMENSIONS

The number of table set containers required is calculated in Equation 21:

$$\begin{aligned} nmr \text{ of table containers} &= 1000 \div ((c L \div set W) \times (c H \div set H)) \\ &= 1000 \div ((12.03 \div 0.4) \times (2.59 \div 1.2)) = 1000 \div (30 \times 2) = 16.67 \approx 17 \text{ containers} \end{aligned} \quad (21)$$

The number of chair stacks containers required was calculated using Equation 22:

$$\begin{aligned} nmr \text{ of table containers} &= 1000 \div ((c L \div stack L) \times (c W \div stack W)) \\ &= 1000 \div ((12.03 \div 0.52) \times (2.44 \div 0.48)) = 1000 \div (23 \times 5) = 8.70 \approx 9 \text{ containers} \end{aligned} \quad (22)$$

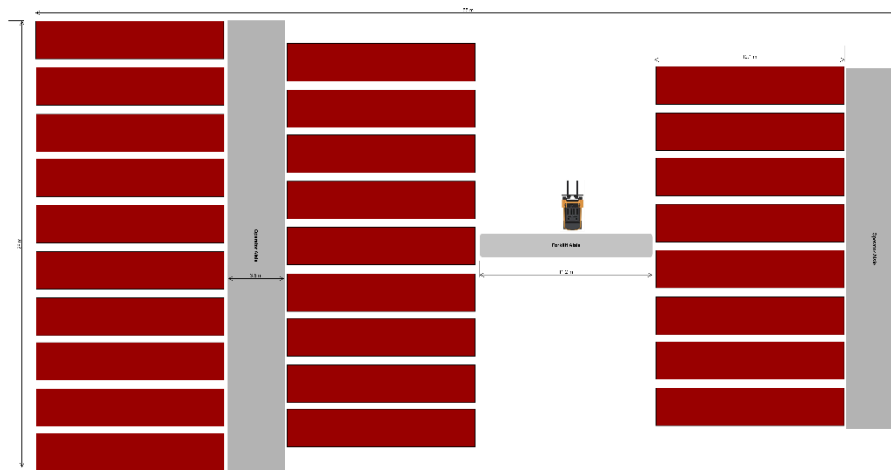


FIGURE 56: FINISHED GOODS STORAGE LAYOUT

Customer Dispatch Layout

- Assume the facility stores one daily production output to sell to customers:
 - 30% of the finished goods are on display or on shelves (75 table sets)
 - 70% are packaged sets to be delivered via truck (35 packaged sets)

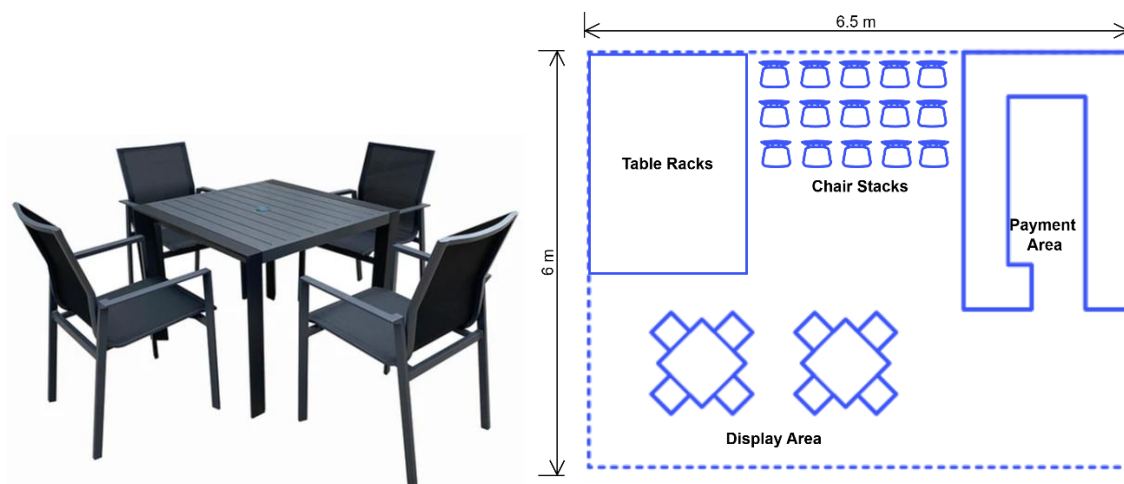


FIGURE 57: RETAIL STORE PRODUCT DISPLAY AND LAYOUT

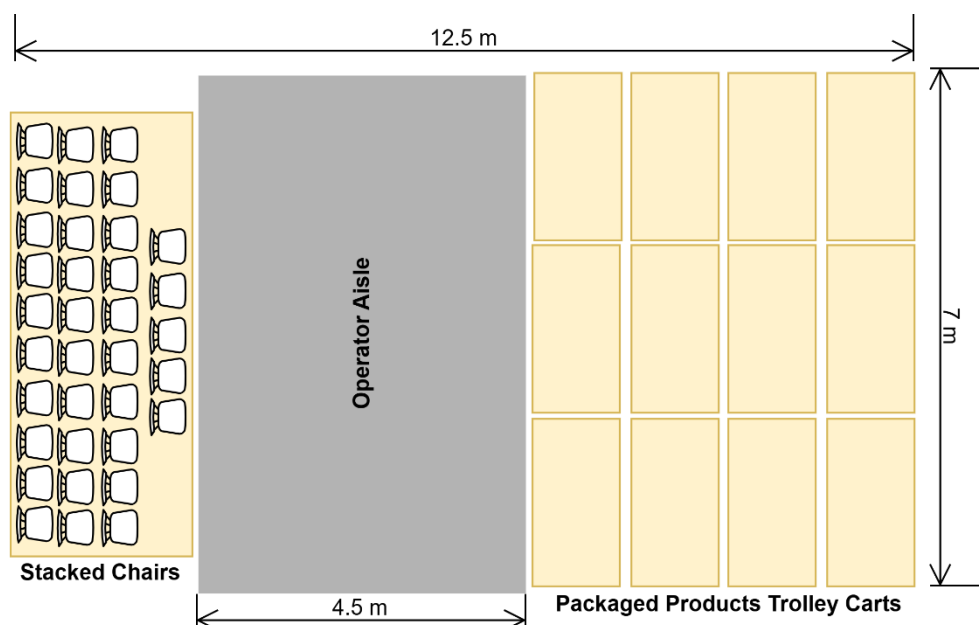


FIGURE 58: LARGE ORDERS STORAGE AREA



FIGURE 59: DELIVERY TRUCK

TABLE 30: DELIVERY TRUCK SPECIFICATIONS

Equipment Specifications	
Dimensions (L x W x H)	9.4m x 2.4m x 2.6m
Operator	1
Equipment cost	R 475 000.00

The cargo truck has a cargo size of 8.5m x 2.4 m x 1.6 m (L x W x H), and has a load capacity of 20 packaged sets as in the configuration in Figure 60:

- 20 table sets (2 box rows)
- 20 chair stacks (5 vertical chair rows)

∴ The truck makes 2 deliveries per day; thus 1 truck is required

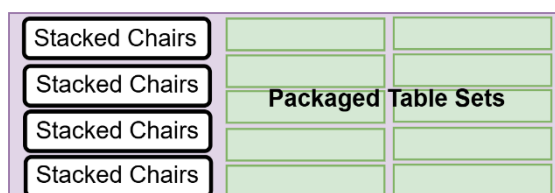


FIGURE 60: DELIVERY TRUCK CONFIGURATION

6.5. Space Requirements

- Assume that the machines and operators have an efficiency level of 80%.

The efficiency level will influence the space requirements, and total machinery required. Equation 23 was applied to all processes (see Table 9 and area calculations) to determine the overall space requirements of the facility as summarised in Table 31.

$$\text{Efficiency factor} = \frac{\text{Production output}}{\text{Production input}} = \frac{100}{80} = 1.25$$

$$\text{Workstation adjustment} = \text{Current workstations} \times \text{efficiency factor} \quad (23)$$

TABLE 31: PRODUCTION FACILITY SPACE REQUIREMENTS

Process	Current Workspace Requirement	Workstation Adjustment (stations)	Area with Efficiency Adjustment (m ²)	Nr of Employees (80% adjustment)
Cutting	2.74	3.43 ≈ 4	409.68	4
Welding (auto)	2.60	3.25 ≈ 4	138.34	4
Welding (manual)	20.83	26.04 ≈ 27	502.85	27
Grinding & Polishing	1.04	1.30 ≈ 2	36.29	2
Powder Coating	1.04	1.30 ≈ 2	1092.00	2
Tabletop Assembly	2.60	3.25 ≈ 4	180.00	4
Rubber Inserts	0.52	0.65 ≈ 1	36.00	1
Packaging	0.52	0.65 ≈ 1	108.00	1
Bending	6.25	7.81 ≈ 8	297.89	8
Stacking	1.04	1.3 ≈ 2	124.80	2
Total			2516.17	55
30% Aisle Space			754.85	-
Final Total Area			3271.02	-

The storage space requirements from section 6.4 (Receiving & Dispatch) were calculated using Equation 20 and summarised in Table 32:

TABLE 32: STORAGE SPACE REQUIREMENTS

Storage Area	Area (m ²)	Nr of Employees
Raw Material Storage	1087.80	1
Rattan Products Storage	595.20	1
Finished Goods Storage	1848.00	1
Retail Store	46.80	1
Large Orders Storage	105.00	1
Total	3682.80	5
30% Aisle Space	1104.84	-
Final Total Area	4787.67	-

The supporting facilities space requirements are summarised in Table 33 as calculated in Appendix B:

TABLE 33: SUPPORTING FACILITIES SPACE REQUIREMENTS

Storage Area	Area (m ²)
Office Space	282.48
Amenities	95.62
Parking	1591.76
Loading Bay	1267.20
Total	3237.06

7. Design Alternatives

The Activity Relationship Chart was partially completed by a shopfloor worker (in black), indicating that the placement of the automated robot welding should not be close to the areas indicated with an X due to safety and ventilation concerns. I completed the rest of the chart in blue and assessed the number of “closeness” ratings between processes. Three alternatives, in the form of block layouts, were designed using this information.

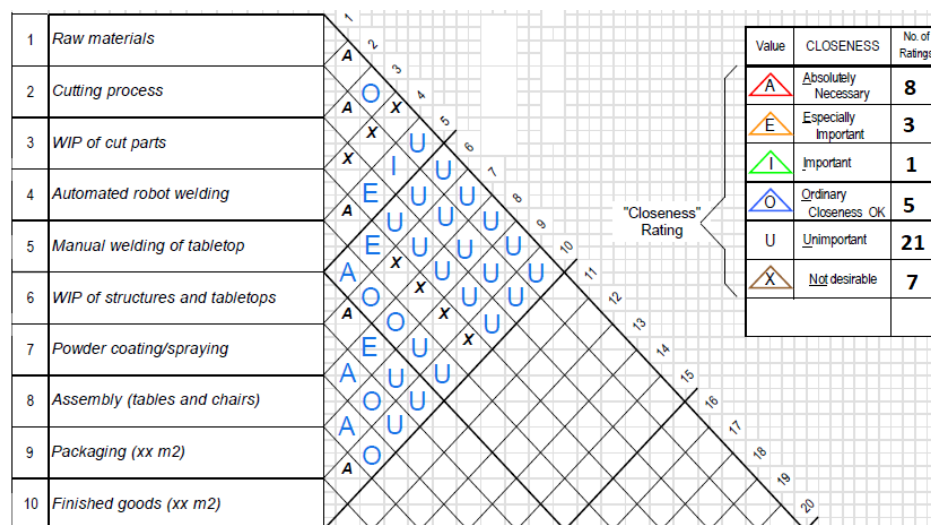
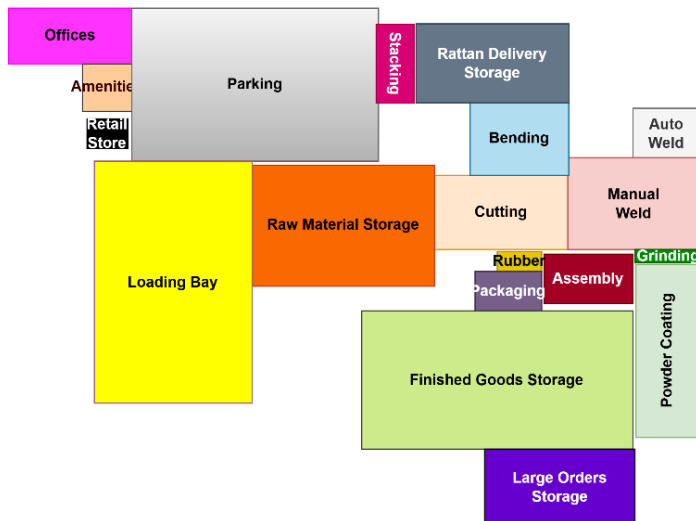
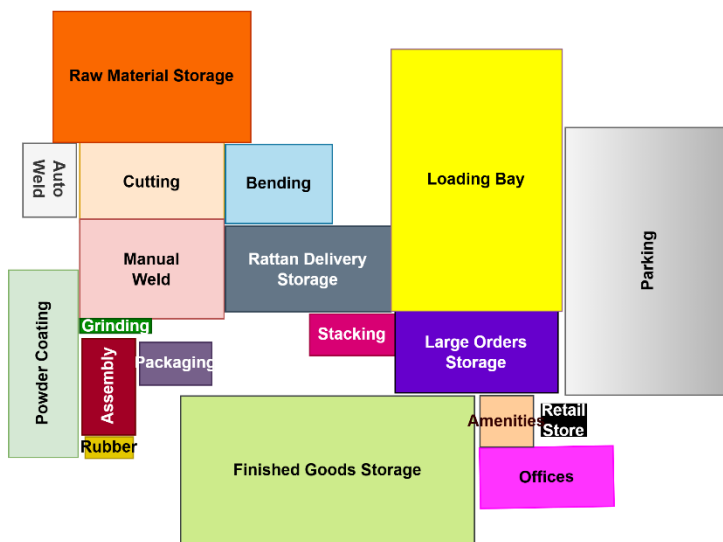


FIGURE 61: ACTIVITY RELATIONSHIP CHART



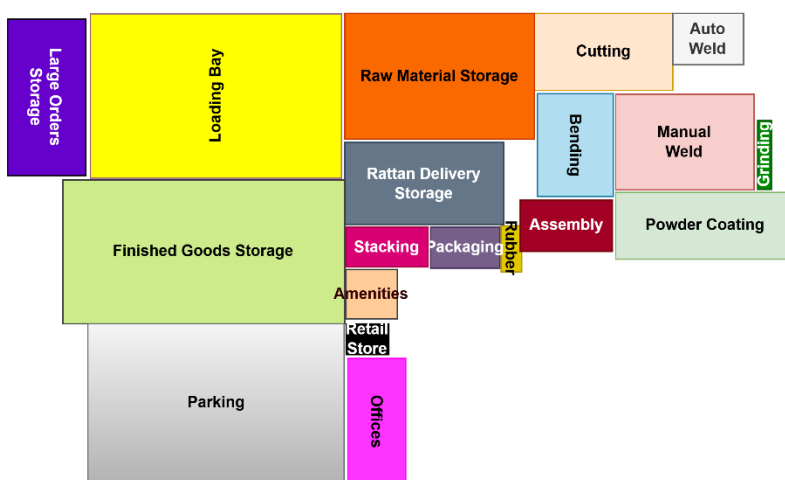
The first alternative was developed by placing the automated welding block first as it has the most restrictions. The layout has a lot of empty spaces, and the loading bay is only close to the raw material storage, and not finished goods, or larger orders.

FIGURE 62: ALTERNATIVE LAYOUT 1



For alternative 2, the focus was on grouping similar processes together (raw material, cutting, welding). However, there is a central open space, and raw material is far from the loading bay.

FIGURE 63: ALTERNATIVE LAYOUT 2



The loading bay was the focus for alternative 3, and the goal was to ensure that the receiving and dispatch activities are close to it. Additionally, the common processes are grouped together and there is minimal wasted space.

FIGURE 64: ALTERNATIVE LAYOUT

7.1. Alternative Evaluation

An acceptance criteria table, Table 34, was used to evaluate the different alternatives. The “best” alternative was chosen based on whether the requirements are met, where the alternative with the highest ranking is considered the best. The scoring is given as, Low=1, Medium=2, and High=3.

TABLE 34: ALTERNATIVE LAYOUTS EVALUATION

Criteria	Alternative 1	Alternative 2	Alternative 3
Total distance travelled (here a lower ranking is better)	High	Med	Low
Space utilisation	Low	Med	High
Effective process flow	Low	High	High
Ease of adding future business	Med	High	Low
Fulfilment of activity chart prerequisites	High	Med	Med
Material Handling	Low	Low	High
Parking in an effective place	Med	Low	Med
Alternative Rankings	12	14	17

Therefore, alternative 3 is the best choice for developing the Master Facility Plan.

8. The Final Design

The final master facility layout was developed in Figure 65 with the legend in appendix C.

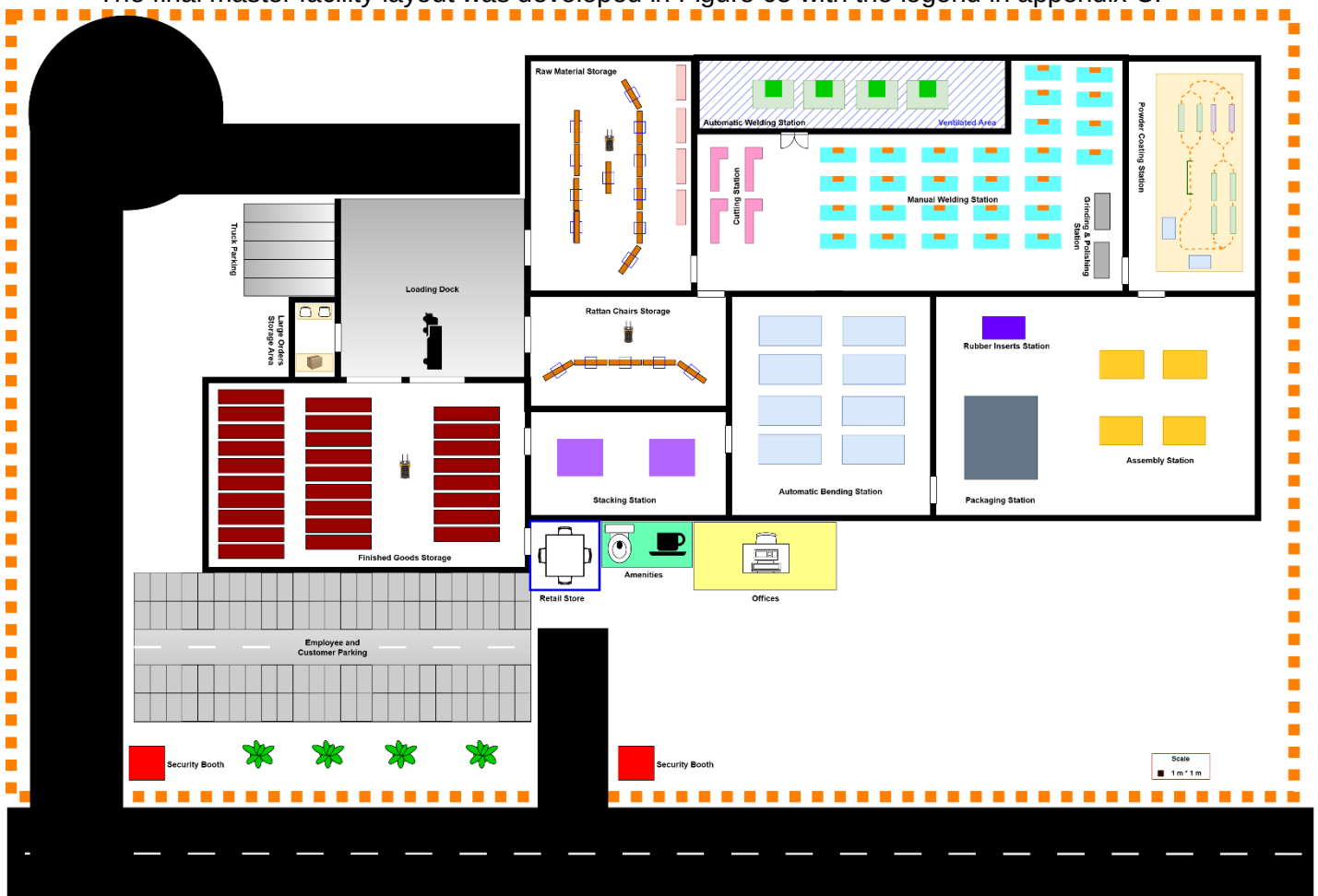


FIGURE 65: THE FINAL MFP LAYOUT

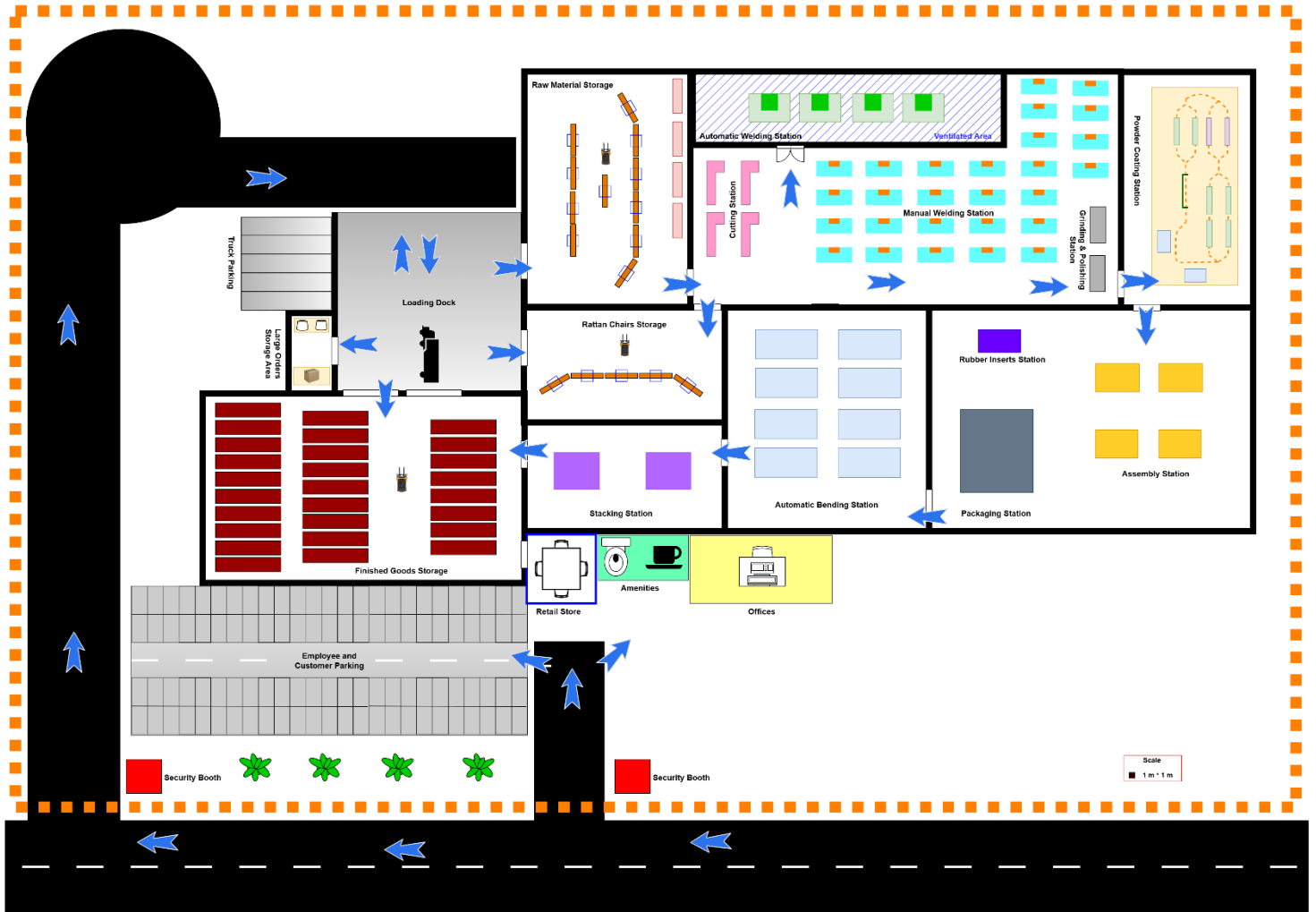


FIGURE 66: MFP MATERIAL AND PEOPLE FLOW

Figure 66 showcases the flow of people and materials throughout the facility.

9. Appendix A

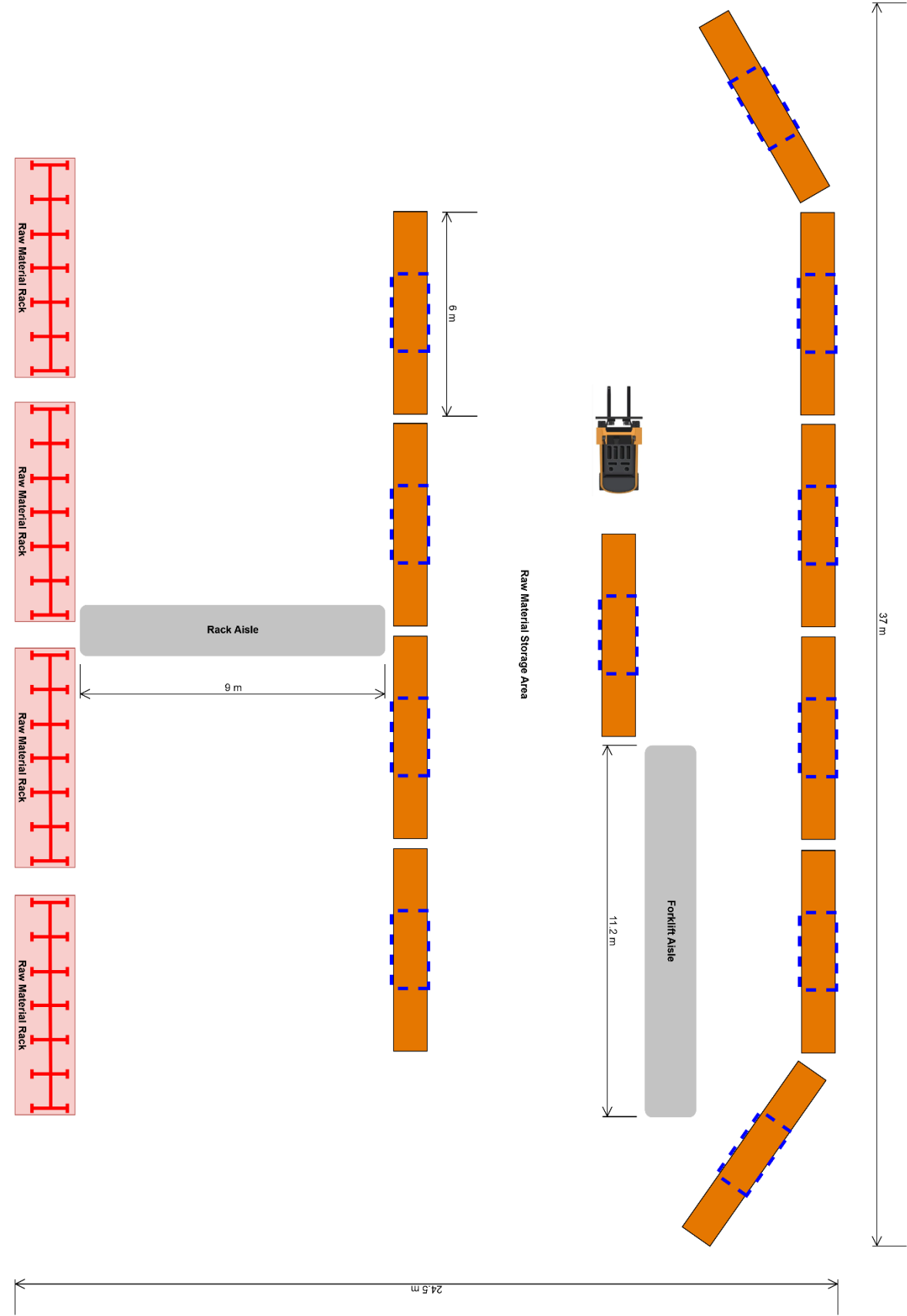


FIGURE 67: ENLARGED RAW MATERIAL STORAGE AREA

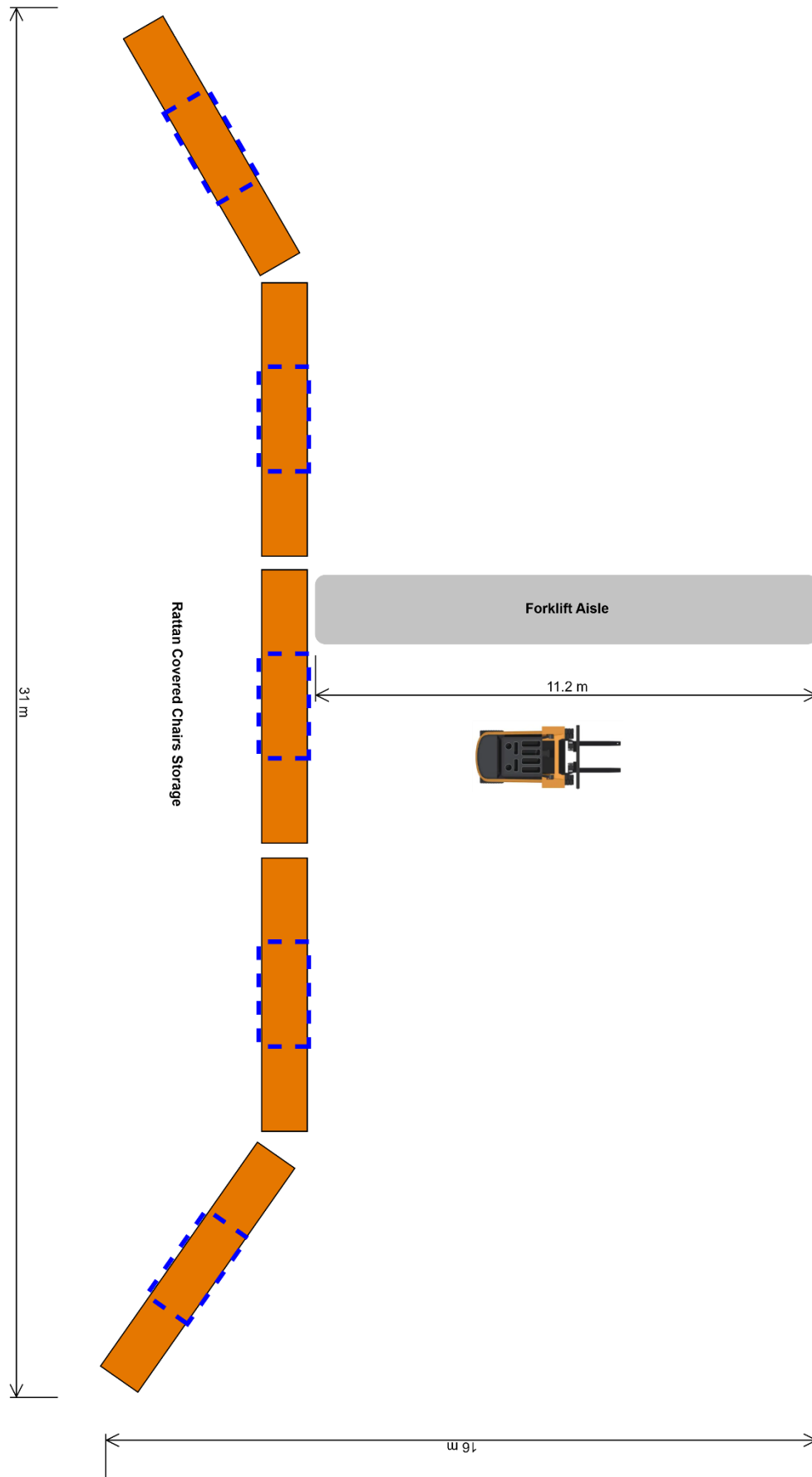


FIGURE 68: ENLARGED RATTAN PRODUCTS LAYOUT

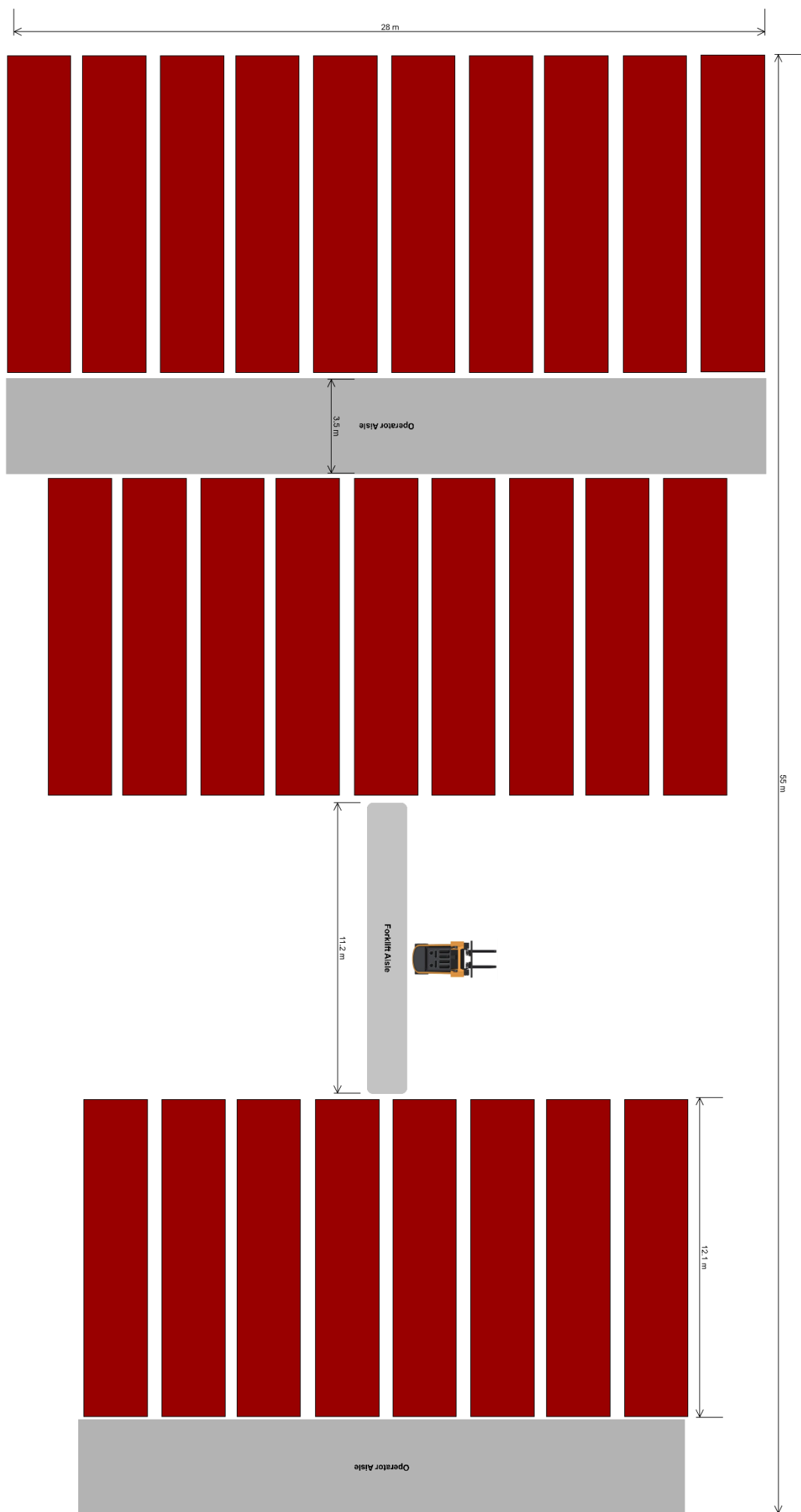


FIGURE 69: ENLARGED FINISHED GOODS STORAGE LAYOUT

10. Appendix B

SUPPORTING STAFF

The office staff and their respective office sizes (based on Equation 20 and Figures 2 - 3) are summarised in Table 35:

TABLE 35: SUPPORTING OFFICE STAFF SPACE REQUIREMENTS

Job Title	Office Size (m^2)	Nmr of Employees
Senior Management	28.50	1
Middle Management	27.98	2
Operations Supervisor	40.50	2
Accountant	15.52	1
Sales Representative	81.00	4
Administrative Staff	79.38	5
Security Personnel	9.60	2
Total	282.48	17

Thus, the total number of employees is calculated as:

$$\begin{aligned}
 \text{total employees} &= \text{production staff} + \text{warehousing staff} + \text{supporting staff} \\
 &= 55 + 5 + 17 = 77 \text{ employees}
 \end{aligned}$$

AMENITIES

The supporting amenities space requirement was calculated using Equation 20 and Figures 5 – 6 and summarised in Table 36:

Assumptions

- There is 1 bathroom stall for every 30 employees (male & female)
- There are separate bathroom facilities for customers (male & female)

$$\text{Employee Bathrooms} = 77 \div 30 \times 2 = 5.13 \approx 6 \text{ bathroom stalls}$$

$$\text{Customer Bathrooms} = 15 \div 30 \times 2 = 1 \approx 2 \text{ bathroom stalls (male \& female)}$$

TABLE 36: AMENITIES SPACE REQUIREMENTS

Facility	Nmr of Facilities	Facility Area (m^2)
Kitchen (3.2m x 3.15m): Office Staff	1	12.10
Kitchen (6.4m x 5m): Production & Warehouse Staff	1	38.40
Employee Bathroom (4.7m x 3m)	2	33.84
Customer Bathroom (4.7m x 1m)	2	11.28
Total	5	95.62

PARKING

The parking spaces required was calculated using Equation 20 and Figures 4 and summarised in Table 37:

Assumptions

- 1 parking space for each employee
- The customer parking is a ration of 1:5, respective to the total employee parking
- There are 5 truck parking spaces

TABLE 37: PARKING SPACE REQUIREMENTS

Parking	Nmr of Parking Spaces	Total Parking Size (m^2)
Employees	77	1100.23
Customers	15	214.33
Trucks	5	277.20
Total	97	1591.76

LOADING BAY

The loading bay area was designed with five loading docks, two docks for raw materials and components receiving, one dock for large orders dispatch (TIPTOP TABLES fleet), and two docks for container loading of export shipments.

$$\begin{aligned}\text{loading dock space requirement} &= \text{nmr loading docks}(\text{dock } L \times \text{dock } W) + 20\% \\ &= 5(32 \times 6.6) + 20\% = 1267.20 \text{ } m^2\end{aligned}$$

11. Appendix C

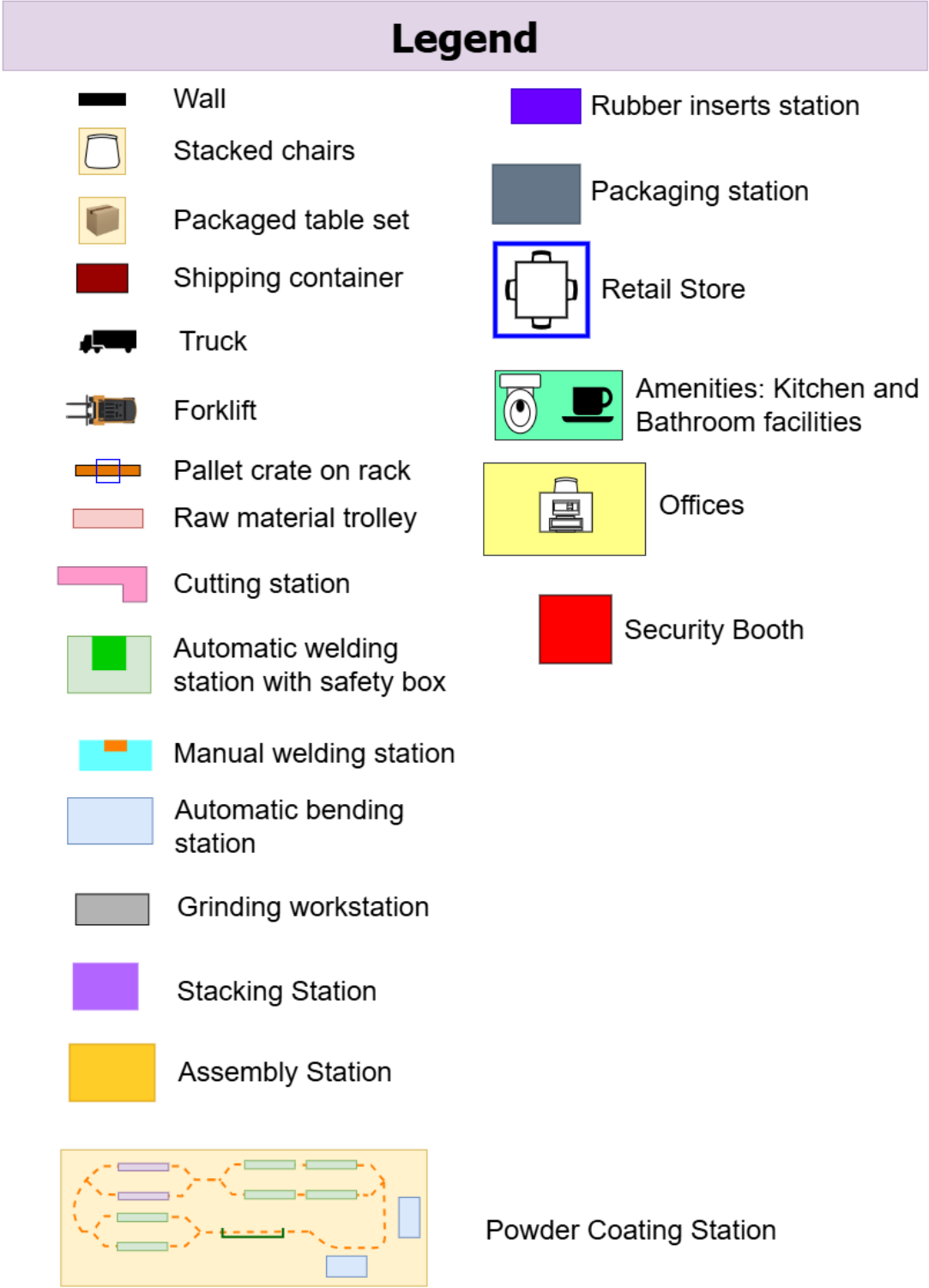


FIGURE 70: THE LEGEND FOR THE MFP

12. References

[1] Aerguvan, "Powder Coating Machine Setup | Powder Coating Equipment," *Powder Coating Equipment | Powder Coating Booth, Powder Curing and Drying Oven, Automatic and Manual Powder Coating Equipment Manufacturer*, Nov. 30, 2022. <https://ems-powdercoating.com/powder-coating-machine-setup/> (accessed Jun. 11, 2025).

[2] "Chair Leg Floor Protector Furniture Table Feet Square Cover Rubber Cap Pads Caps," *eBay Australia*, 2025. <https://www.ebay.com.au/itm/354457779294> (accessed Jun. 11, 2025).